第三章 一元函数积分学

§1 定积分的概念、性质和微积分基本定理

习题. P.119.

1. 根据定积分定义，计算下列积分：

(1) $\int_a^b x\,dx$.

解: $\int_a^b x\,dx = \lim\limits_{n\to\infty}\sum\limits_{i=1}^{n}\left(a+\frac{b-a}{n}\cdot i\right)\frac{b-a}{n} = \lim\limits_{n\to\infty}\left(na+\frac{(b-a)^{\bullet}}{n^{\bullet}}\cdot\frac{n(n+1)}{2}\right)\cdot\frac{b-a}{n}$

$= \lim\limits_{n\to\infty}\left(a(b-a)+\frac{(b-a)^2}{2}\cdot\frac{n+1}{n}\right) = \frac{1}{2}(b^2-a^2)$.

(2) $\int_0^1 e^x\,dx$

解: $\int_0^1 e^x\,dx = \lim\limits_{n\to\infty}\sum\limits_{i=1}^{n} e^{\frac{i}{n}}\cdot\frac{1}{n} = \lim\limits_{n\to\infty}\frac{1}{n}\cdot\frac{e^{\frac{1}{n}}(1-e^{\frac{n}{n}})}{1-e^{\frac{1}{n}}} = (1-e)\lim\limits_{n\to\infty}\frac{e^{\frac{1}{n}}}{n(1-e^{\frac{1}{n}})}$

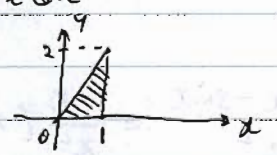
而 $\lim\limits_{n\to\infty} n(1-e^{\frac{1}{n}}) = \lim\limits_{x\to 0}\frac{1}{x}(1-e^x) = \lim\limits_{x\to 0}\frac{1-e^x}{x} = \lim\limits_{x\to 0}(-e^x) = -1$.

$\therefore \int_0^1 e^x\,dx = (1-e)\cdot\lim\limits_{n\to\infty}\frac{e^{\frac{1}{n}}}{n(1-e^{\frac{1}{n}})} = (1-e)\cdot(-1) = e-1$.

2. 利用定积分的几何意义，计算下列积分：

(1) $\int_0^1 2x\,dx$

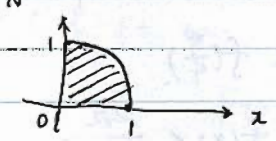
解:



$\int_0^1 2x\,dx = S = \frac{1}{2}\cdot 1\cdot 2 = 1$.

(2) $\int_0^1 \sqrt{1-x^2}\,dx$

解:



$\int_0^1 \sqrt{1-x^2}\,dx = S = \frac{1}{4}\pi\cdot 1^2 = \frac{\pi}{4}$.

3. 比较积分的大小：

(1) $\int_0^1 x\,dx$ 和 $\int_0^1 \sqrt{x}\,dx$

解: $\forall x\in(0,1)$, $x<\sqrt{x}$. $\therefore \int_0^1 x\,dx < \int_0^1 \sqrt{x}\,dx$.

(2) $\int_0^{\frac{\pi}{2}} x\,dx$ 和 $\int_0^{\frac{\pi}{2}} \sin x\,dx$.

解: $\forall x \in (0, \frac{\pi}{2})$, $x > \sin x \Rightarrow \int_0^{\frac{\pi}{2}} x\,dx > \int_0^{\frac{\pi}{2}} \sin x\,dx$.

(3) $\int_0^1 x\,dx$ 和 $\int_0^1 \ln(1+x)\,dx$

解: 令 $f(x) = x - \ln(1+x)$. $f'(x) = 1 - \frac{1}{1+x} > 0$. $x \in (0,1)$.

$\therefore f(x) > f(0) = 0 \Rightarrow x > \ln(1+x) \Rightarrow \int_0^1 x\,dx > \int_0^1 \ln(1+x)\,dx$.

(4) $\int_0^1 e^x\,dx$ 和 $\int_0^1 (1+x)\,dx$

解: 令 $f(x) = e^x - (1+x)$. $f'(x) = e^x - 1 > 0 \quad \forall x \in (0,1)$

$\therefore f(x) > f(0) = 0 \Rightarrow e^x > 1+x \Rightarrow \int_0^1 e^x\,dx > \int_0^1 (1+x)\,dx$.

4. 设 f 是 $[a,b]$ 上非负连续函数, 且 $f(x) \neq 0$. 证明: $\int_a^b f(x)\,dx > 0$.

证明: 令 $F(x) = \int_a^x f(t)\,dt$. 则 $F'(x) = f(x) \geq 0$. $F(x)$ 为单调递增函数.

由 $b > a \Rightarrow F(b) > F(a)$

$\therefore \int_a^b f(x)\,dx = F(b) - F(a) > 0$.

5. 证明不等式

(1) $\frac{1}{2} < \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\sin x}{x}\,dx < \frac{\sqrt{2}}{2}$.

证明: 令 $f(x) = \frac{\sin x}{x}$. $x \in (\frac{\pi}{4}, \frac{\pi}{2})$. $f'(x) = \frac{x\cos x - \sin x}{x^2} = \frac{\cos x}{x^2}(x - \tan x)$.

$\because \sin x < x < \tan x$, $\forall x \in (0, \frac{\pi}{2})$. $\therefore f'(x) < 0$. $x \in (\frac{\pi}{4}, \frac{\pi}{2})$.

$f(x)$ 在 $(\frac{\pi}{4}, \frac{\pi}{2})$ 上单调递减. $\Rightarrow f(\frac{\pi}{2}) < f(x) < f(\frac{\pi}{4})$.

$f(\frac{\pi}{2}) = \frac{2}{\pi}$. $f(\frac{\pi}{4}) = \frac{4}{\pi} \cdot \frac{\sqrt{2}}{2} = \frac{2\sqrt{2}}{\pi}$. 即: $\frac{2}{\pi} < f(x) < \frac{2\sqrt{2}}{\pi}$.

(两边从 $\frac{\pi}{4}$ 到 $\frac{\pi}{2}$ 积分): $\frac{2}{\pi} \cdot \frac{\pi}{4} < \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} f(x)\,dx < \frac{2\sqrt{2}}{\pi} \cdot \frac{\pi}{4}$

即: $\frac{1}{2} < \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} f(x)\,dx < \frac{\sqrt{2}}{2}$.

(2) $\frac{2}{5} < \int_1^2 \frac{x}{1+x^2}\,dx < \frac{1}{2}$.

证明: 令 $f(x) = \frac{x}{1+x^2}$. $f'(x) = \frac{1+x^2 - 2x^2}{1+x^2} = \frac{1-x^2}{1+x^2} < 0$. $x \in (1,2)$.

∴ $f(x)$ 在 $(1,2)$ 上单调递减，$f(2) < f(x) < f(1)$

两边从1到2积分，得：$\frac{2}{5} < \int_1^2 \frac{x}{1+x^2}dx < \frac{1}{2}$.

6. 计算下列导数：

(1) $\frac{d}{dx}\int_0^{x^2}\sqrt{1+t^4}\,dt = \sqrt{1+(x^2)^4}\cdot 2x = 2x\sqrt{1+x^8}$.

(2) $\frac{d}{dx}\int_x^{\sin x}\frac{dt}{\sqrt{1+t^2}} = \frac{\cos x}{\sqrt{1+(\sin x)^2}} - \frac{1}{\sqrt{1+x^2}} = \frac{\cos x}{\sqrt{1+\sin^2 x}} - \frac{1}{\sqrt{1+x^2}}$.

7. 求下列极限：

(1) $\lim_{x\to 0}\frac{\int_0^x \cos t^2 dt}{x} = \lim_{x\to 0}\frac{\frac{d}{dx}\int_0^x \cos t^2 dt}{1} = \lim_{x\to 0}\frac{\cos x^2}{1} = 1$.

(2) $\lim_{x\to 0}\frac{\int_{\cos x}^1 e^{-t^2}dt}{x^2} = \lim_{x\to 0}\frac{-e^{-\cos^2 x}\cdot(-\sin x)}{2x} = \lim_{x\to 0}\frac{\sin x}{x}\cdot\frac{1}{2}e^{-\cos^2 x} = \frac{1}{2e}$.

8. 设 $F(x) = \int_0^x (t^2-x^2)f'(t)\,dt$，求 $F'(x)$.

解：$F(x) = \int_0^x (t^2-x^2)f'(t)dt = \int_0^x t^2 f'(t)dt - x^2\int_0^x f'(t)dt$

$F'(x) = x^2 f'(x) - 2x\int_0^x f'(t)dt - x^2 f'(x) = -2x\int_0^x d\,f(t)$

$= -2x\left[f(t)\right]_0^x = -2x\left[f(x)-f(0)\right]$.

9. 设 f 是 $(0,+\infty)$ 上的正值连续函数，证明：

$F(x) = \frac{\int_0^x t f(t)dt}{\int_0^x f(t)dt}$ 是 $(0,+\infty)$ 内的单调增加函数.

证明：$F'(x) = \frac{\frac{d}{dx}\int_0^x t f(t)dt\cdot\int_0^x f(t)dt - \int_0^x t f(t)dt\cdot\frac{d}{dx}\int_0^x f(t)dt}{\left(\int_0^x f(t)dt\right)^2}$

$= \frac{x f(x)\cdot\int_0^x f(t)dt - f(x)\int_0^x t f(t)dt}{\left(\int_0^x f(t)dt\right)^2} = \frac{f(x)\cdot\int_0^x (x-t)f(t)dt}{\left(\int_0^x f(t)dt\right)^2}$

$\forall x \in (0,+\infty)$. 已知 $f(x)>0$. $\forall 0<t<x$: $(x-t)f(t)>0$.

$\therefore \int_0^x (x-t)f(t)\,dt > 0$. $\forall x \in (0,+\infty)$.

$\therefore F'(x)>0 \Rightarrow F(x)$ 在 $(0,+\infty)$ 内单调增加.

10. 设 f 在 $[a,b]$ 连续, 在 (a,b) 可导, 且 $f'(x)\le 0$.

$$F(x)=\frac{1}{x-a}\int_a^x f(t)\,dt$$

证明: 在 (a,b) 内有 $F'(x)\le 0$.

证明: $F'(x)=-\frac{1}{(x-a)^2}\int_a^x f(t)\,dt+\frac{1}{x-a}\cdot f(x)=\frac{1}{(x-a)^2}\left[(x-a)f(x)-\int_a^x f(t)\,dt\right]$.

$\le$ $g(x)=(x-a)f(x)-\int_a^x f(t)\,dt$. $g(a)=0$.

$g'(x)=f(x)+(x-a)f'(x)-f(x)=(x-a)f'(x)\le 0$. $\forall x\in(a,b)$.

即 $g(x)$ 在 (a,b) 内单调递减. $\therefore g(x)<g(a)=0$.

$\therefore F'(x)=\frac{g(x)}{(x-a)^2}\le 0$. $\forall x\in(a,b)$.

11. 计算下列定积分:

(1) $\int_0^{\frac{\pi}{4}} \cos x\,dx = \sin x\Big|_0^{\frac{\pi}{4}}=\frac{\sqrt{2}}{2}$.

(2) $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sec x\tan x\,dx = \sec x\Big|_{-\frac{\pi}{4}}^{\frac{\pi}{4}}=0$.

(3) $\int_{\frac{1}{2}}^{\frac{\sqrt{2}}{2}} \frac{dx}{\sqrt{1-x^2}} = \arcsin x\Big|_{\frac{1}{2}}^{\frac{\sqrt{2}}{2}}=\frac{\pi}{3}-\frac{\pi}{6}=\frac{\pi}{6}$.

(4) $\int_2^3 (x+1)e^x\,dx = (xe^x)\Big|_2^3=3e^3-2e^2$.

12. 设函数在 $[a,b]$ 连续, 在 (a,b) 内可导, 且满足 $\frac{2}{b-a}\int_a^{\frac{a+b}{2}} f(x)\,dx=f(b)$.

证明: 存在 $\xi\in(a,b)$, 使得 $f'(\xi)=0$.

证明: $\because f(x)$ 在 $[a,b]$ 连续, $[a,\frac{a+b}{2}]\subset[a,b]$. $\therefore \exists \eta\in[a,\frac{a+b}{2})$, 成立.

$$f(\eta)=\frac{2}{b-a}\int_a^{\frac{a+b}{2}} f(x)\,dx=f(b).$$

∵ $f(x)$ 在 $[\eta, b] \subset [a,b]$ 上连续，在 (η, b) 可导，$f(\eta) = f(b)$.
由 Rolle 定理，$\exists\, \xi \in (\eta, b) \subset (a,b)$，使得 $f'(\xi) = 0$.

13. 设 $f(x)$ 在 $[0,1]$ 连续，且单调减少，证明对任意 $\alpha \in [0,1]$，成立：

$$\int_0^\alpha f(x)\,dx \geq \alpha \int_0^1 f(x)\,dx.$$

证明：$\alpha \int_0^1 f(x)\,dx = \alpha \left(\int_0^\alpha f(x)\,dx + \int_\alpha^1 f(x)\,dx \right)$ $\alpha \in [0,1]$.

考虑积分 $(1-\alpha) \int_0^\alpha f(x)\,dx \;?\; \alpha \int_\alpha^1 f(x)\,dx$.

由积分中值定理：$\exists\, x_1 \in [0,\alpha]$ 和 $x_2 \in [\alpha, 1]$，使得：

$$(1-\alpha) \int_0^\alpha f(x)\,dx = \alpha(1-\alpha) f(x_1), \qquad x_1 \in [0,\alpha].$$

$$\alpha \int_\alpha^1 f(x)\,dx = \alpha(1-\alpha) f(x_2), \qquad x_2 \in [\alpha, 1].$$

∵ $f(x)$ 在 $[0,1]$ 单调减少，$x_1 \leq x_2 \Rightarrow f(x_1) \geq f(x_2)$.

∴ $(1-\alpha) \int_0^\alpha f(x)\,dx \geq \alpha \int_\alpha^1 f(x)\,dx$.

即：$\int_0^\alpha f(x)\,dx \geq \alpha \int_0^1 f(x)\,dx$.

14. 设 $f(x)$ 在 $[0,a]$ 上二阶可导 $(a>0)$，且 $f''(x) \geq 0$，证明：

$$\int_0^a f(x)\,dx \geq a f\left(\frac{a}{2}\right).$$

证明：将 $f(x)$ 在 $x = \frac{a}{2}$ 处展开成一阶泰勒公式：

$$f(x) = f\left(\frac{a}{2}\right) + f'\left(\frac{a}{2}\right)\left(x - \frac{a}{2}\right) + \frac{1}{2} f''(\xi)\left(x - \frac{a}{2}\right)^2, \qquad \xi \in (0,a).$$

∵ $f''(x) \geq 0$. ∴ $f(x) \geq f\left(\frac{a}{2}\right) + f'\left(\frac{a}{2}\right)\left(x - \frac{a}{2}\right)$.

两边从 0 到 a 积分，有：

$$\int_0^a f(x)\,dx \geq \int_0^a f\left(\frac{a}{2}\right) dx + \int_0^a f'\left(\frac{a}{2}\right)\left(x - \frac{a}{2}\right) dx$$

$$= a f\left(\frac{a}{2}\right) + f'\left(\frac{a}{2}\right) \cdot \left(\frac{1}{2}x^2 - \frac{a}{2}x\right)\Big|_0^a = a f\left(\frac{a}{2}\right).$$

∴ $\int_0^a f(x)\,dx \geq a f\left(\frac{a}{2}\right)$.

§2. 不定积分的计算

习题. P.138.

1. 计算下列不定积分:

(1) $\int \sqrt[n]{x^m}\,dx = \int x^{\frac{m}{n}}\,dx = \frac{1}{1+\frac{m}{n}} x^{1+\frac{m}{n}} + C = \frac{n}{n+m} x^{\frac{n+m}{n}} + C.$

(2) $\int (x^2 - x^{-2})\sqrt{x\sqrt{x}}\,dx = \int \left(x^{\frac{11}{4}} - x^{-\frac{5}{4}}\right)dx = \frac{4}{15}x^{\frac{15}{4}} + 4x^{-\frac{1}{4}} + C$

(3) $\int \frac{x^2-1}{x^2+1}\,dx = \int \left(1 - \frac{2}{1+x^2}\right)dx = x - 2\arctan x + C.$

(4) $\int e^x (x^2 e^{-x} + 2)\,dx = \int (x^2 + 2e^x)\,dx = \frac{1}{3}x^3 + 2e^x + C.$

(4) $\int 3^x e^x\,dx = \int (3e)^x\,dx = \frac{1}{\ln(3e)}\cdot (3e)^x + C = \frac{1}{\ln 3 + 1}\cdot 3^x e^x + C.$

(5) $\int \left(\cot^2 x + \frac{3}{\sqrt{1-x^2}}\right)dx = \int \frac{\cos^2 x}{\sin^2 x}\,dx + 3\arcsin x = \int (\csc^2 x - 1)\,dx + 3\arcsin x$

$= -\cot x - x + 3\arcsin x + C.$

(7) $\int (3\operatorname{sh} x + 4\operatorname{ch} x)\,dx = 3\operatorname{ch} x + 4\operatorname{sh} x + C.$

(8) $\int \frac{1}{\sin^2 x \cos^2 x}\,dx = \int \frac{4}{(\sin 2x)^2}\,dx = -4\int \csc^2 2x\,dx = 2\int \csc^2 2x\,d(2x) = -\cot 2x + C.$

$\left(= \int \frac{\sin^2 x + \cos^2 x}{\sin^2 x \cos^2 x}\,dx = \int (\sec^2 x + \csc^2 x)\,dx = \tan x - \cot x + C. \right)$

2. 计算下列不定积分:

(1) $\int \frac{dx}{x^2+2} = \frac{1}{\sqrt{2}}\arctan\frac{x}{\sqrt{2}} + C.$

(2) $\int \frac{10x^2}{(x^3+1)^2}\,dx = \frac{10}{3}\int \frac{d(x^3+1)}{(x^3+1)^2} = -\frac{10}{3(x^3+1)} + C.$

(3) $\int \frac{x\,dx}{\sqrt{4x^2+9}} = \frac{1}{8}\int \frac{d(4x^2+9)}{\sqrt{4x^2+9}} = \frac{1}{4}\sqrt{4x^2+9} + C.$

(4) $\int \frac{2x+3x^2}{4+x^2+x^3}\,dx = \int \frac{d(x^3+x^2+4)}{x^3+x^2+4} = \ln|x^3+x^2+4| + C.$

(5) $\int \frac{e^x}{\sqrt{1+2e^x}}\,dx = \frac{1}{2}\int \frac{d(1+2e^x)}{\sqrt{1+2e^x}} = \sqrt{1+2e^x} + C.$

(6) $\int x(e^{x^2}+2)\,dx = \frac{1}{2}\int (e^{x^2}+2)\,dx^2 = \frac{1}{2}e^{x^2} + x^2 + C.$

(7) $\int x^{-2} e^{\frac{1}{x}}\,dx = -\int e^{\frac{1}{x}}\,d\left(\frac{1}{x}\right) = -e^{\frac{1}{x}} + C.$

(8) $\int \frac{dx}{e^x + e^{-x}} = \int \frac{e^x}{1+e^{2x}}\,dx = \int \frac{d\,e^x}{1+(e^x)^2} = \arctan e^x + C.$

(9) $\int \frac{x\,dx}{\sqrt{4-25x^2}} = -\frac{1}{50}\int \frac{d(4-25x^2)}{\sqrt{4-25x^2}} = -\frac{1}{25}\sqrt{4-25x^2} + C.$

(10) $\int \frac{dx}{x\ln x \ln(\ln x)} = \int \frac{d\ln x}{\ln x \ln(\ln x)} = \int \frac{d\ln(\ln x)}{\ln(\ln x)} = \ln\ln\ln x + C.$

(11) $\int \frac{2^x}{\sqrt{4-4^x}}\,dx = \frac{1}{\ln 2}\int \frac{1}{\sqrt{4-(2^x)^2}}\,d(2^x) = \frac{\arcsin\frac{2^x}{2}}{\ln 2} + C$

$= \frac{1}{\ln 2}\arcsin 2^{x-1} + C.$

(12) $\int \frac{\arctan x}{1+x^2}\,dx = \int \arctan x\, d(\arctan x) = \frac{1}{2}(\arctan x)^2 + C.$

(13) $\int \sqrt{\frac{\arcsin x}{1-x^2}}\,dx = \int \sqrt{\arcsin x}\, d(\arcsin x) = \frac{2}{3}(\arcsin x)^{\frac{3}{2}} + C.$

(14) $\int \frac{x^2}{3-x^2}\,dx = \int\left(\frac{3}{3-x^2} - 1\right)dx = 3\cdot\frac{1}{2\sqrt{3}}\ln\left|\frac{\sqrt{3}+x}{\sqrt{3}-x}\right| - x + C$

$= \frac{\sqrt{3}}{2}\ln\left|\frac{\sqrt{3}+x}{\sqrt{3}-x}\right| - x + C.$

(15) $\int \frac{x\,dx}{x^2+x-6} = \int \frac{x\,dx}{(x+3)(x-2)} = \int\left(\frac{3}{5}\cdot\frac{1}{x+3} + \frac{2}{5}\cdot\frac{1}{x-2}\right)dx$

$= \frac{3}{5}\ln|x+3| + \frac{2}{5}\ln|x-2| + C.$

(16) $\int \frac{x+1}{\sqrt{3+2x-x^2}}\,dx = \int \frac{2}{\sqrt{3+2x-x^2}}\,dx - \frac{1}{2}\int \frac{d(3+2x-x^2)}{\sqrt{3+2x-x^2}}$

$= \int \frac{2}{\sqrt{4-(x-1)^2}}\,d(x-1) - \frac{1}{2}\int \frac{d(3+2x-x^2)}{\sqrt{3+2x-x^2}}$

$= 2\arcsin\frac{x-1}{2} - \sqrt{3+2x-x^2} + C.$

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3. 计算下列不定积分：

(1) $\int x\sqrt{x+1}\,dx$ 令 $t=\sqrt{x+1}$，$x=t^2-1$，$dx=2t\,dt$

$=\int (t^2-1)\cdot t\cdot 2t\,dt=\int (2t^4-2t^2)\,dt=\frac{2}{5}t^5-\frac{2}{3}t^3+C=\frac{2}{5}(x+1)^{\frac{5}{2}}-\frac{2}{3}(x+1)^{\frac{3}{2}}+C$.

(2) $\int x\sqrt[3]{x+1}\,dx$，令 $t=\sqrt[3]{x+1}$，$x=t^3-1$，$dx=3t^2dt$

$=\int (t^3-1)\cdot t\cdot 3t^2\,dt=\int (3t^6-3t^3)\,dt=\frac{3}{7}t^7-\frac{3}{4}t^4+C=\frac{3}{7}(x+1)^{\frac{7}{3}}-\frac{3}{4}(x+1)^{\frac{4}{3}}+C$.

(3) $\int\sqrt{1-e^x}\,dx$，令 $t=\sqrt{1-e^x}$，$x=\ln(1-t^2)$，$dx=\frac{-2t}{1-t^2}dt$.

$=\int t\cdot\frac{-2t}{1-t^2}dt=2\int\left(\frac{1}{1-t^2}-1\right)dt=2\int\left(\frac{1}{1-t}-\frac{1}{1+t}-1\right)dt$

$=2t+\ln(1-t)-\ln(1+t)+C=2\sqrt{1-e^x}+\ln\frac{1-\sqrt{1-e^x}}{1+\sqrt{1-e^x}}+C$.

(4) $\int\frac{dx}{\sqrt{1+e^x}}$，令 $t=\sqrt{1+e^x}$，$x=\ln(t^2-1)$，$dx=\frac{2t}{t^2-1}dt$

$=\int\frac{2t}{t^2-1}\cdot\frac{1}{t}\,dt=\int\frac{2}{t^2-1}dt=\int\left(\frac{1}{t-1}-\frac{1}{t+1}\right)dt$

$=\ln(t-1)-\ln(t+1)+C=\ln\frac{\sqrt{1+e^x}-1}{\sqrt{1+e^x}+1}+C$.

(5) $\int\frac{\sqrt{x^2-1}}{x}dx$，令 $x=\sec\theta$，$\sqrt{x^2-1}=\tan\theta$，$dx=\sec\theta\tan\theta\,d\theta$

$=\int\frac{\tan\theta}{\sec\theta}\cdot\sec\theta\tan\theta\,d\theta=\int\tan^2\theta\,d\theta=\int(\sec^2\theta-1)\,d\theta=\tan\theta-\theta+C$

$=\sqrt{x^2-1}-\arccos\frac{1}{x}+C$.

(6) $\int\frac{dx}{x\sqrt{4x^2+9}}=\int\frac{dx}{x^2\sqrt{4+\frac{9}{x^2}}}=-\frac{1}{3}\int\frac{d\left(\frac{3}{x}\right)}{\sqrt{4+\left(\frac{3}{x}\right)^2}}=-\frac{1}{3}\ln\left(\frac{3}{x}+\sqrt{4+\frac{9}{x^2}}\right)+C$

$=\frac{1}{3}\ln x-\frac{1}{3}\ln\left(\sqrt{4x^2+9}+3\right)+C$.

(7) $\int\frac{dx}{x\sqrt{a^2-x^2}}=\int\frac{dx}{x^2\sqrt{\left(\frac{a}{x}\right)^2-1}}=-\frac{1}{a}\int\frac{d\left(\frac{a}{x}\right)}{\sqrt{\left(\frac{a}{x}\right)^2-1}}=-\frac{1}{a}\ln\left|\frac{a}{x}+\sqrt{\left(\frac{a}{x}\right)^2-1}\right|+C$

$=-\frac{1}{a}\ln\left|a+\sqrt{a^2-x^2}\right|+\frac{1}{a}\ln|x|+C$.

(8) $\int \frac{dx}{(1+x^2)^{3/2}}$ 令 $x=\tan\theta$. $dx=\sec^2\theta d\theta$. $1+x^2=\sec^2\theta$.

$$=\int \frac{\sec^2\theta\, d\theta}{\sec^3\theta}=\int \cos\theta\, d\theta=\sin\theta+C=\frac{x}{\sqrt{1+x^2}}+C.$$

(9) $\int \frac{dx}{1+\sqrt{1+x}}$. 令 $t=\sqrt{1+x}$. $x=t^2-1$ $dx=+2t\,dt$

$$=\int \frac{+2t\,dt}{1+t}=\int\left(-2+\frac{2}{1+t}\right)dt=+2t-2\ln|1+t|+C$$

$$=2\sqrt{1+x}-2\ln\left|1+\sqrt{1+x}\right|+C.$$

(10) $\int \frac{\sqrt{x}}{\sqrt{x}-\sqrt[3]{x}}\,dx$. 令 $x^{\frac{1}{6}}=t$. $x=t^6$. $dx=6t^5dt$.

$$=\int \frac{x^{\frac{1}{2}}}{x^{\frac{1}{2}}-x^{\frac{1}{3}}}dx=\int \frac{t^3}{t^3-t^2}\cdot 6t^5\,dt=\int \frac{6t^6}{t-1}dt=6\int\left(\frac{t^6-1}{t-1}+\frac{1}{t-1}\right)dt$$

$$=6\int\left(t^5+t^4+t^3+t^2+t+1+\frac{1}{t-1}\right)dt=t^6+\frac{6}{5}t^5+\frac{3}{2}t^4+2t^3+3t^2+6t+6\ln|t-1|+C$$

$$=x+\frac{6}{5}x^{\frac{5}{6}}+\frac{3}{2}x^{\frac{2}{3}}+2x^{\frac{1}{2}}+3x^{\frac{1}{3}}+6x^{\frac{1}{6}}+6\ln(x^{\frac{1}{6}}-1)+C.$$

4. 计算下列不定积分:

(1) $\int \cos^4 x\,dx=\int \frac{1}{4}(1+\cos 2x)^2dx=\int\left(\frac{1}{4}+\frac{1}{2}\cos 2x+\frac{1}{4}\cos^2 2x\right)dx$

$$=\int\left(\frac{1}{4}+\frac{1}{2}\cos 2x+\frac{1}{8}+\frac{1}{8}\cos 4x\right)dx$$

$$=\frac{3}{8}x+\frac{1}{4}\sin 2x+\frac{1}{32}\sin 4x+C.$$

(2) $\int \tan^3 x\,dx=\int(\sec^2x-1)\cdot\frac{\sin x}{\cos x}dx=-\int\left(\frac{1}{\cos^2 x}-\frac{1}{\cos x}\right)d\cos x$

$$=\frac{1}{2}\sec^2x+\ln|\cos x|+C=\frac{1}{2}\tan^2x-\frac{1}{2}\ln(1+\tan^2x)+C.$$

(3) $\int \frac{dx}{1+\sin x}=\int \frac{1-\sin x}{\cos^2 x}dx=\int\left(\sec^2x-\frac{\sin x}{\cos^2 x}\right)dx$

$$=\tan x+\int \frac{d\cos x}{\cos^2 x}=\tan x-\frac{1}{\cos x}+C$$

(4) $\int \frac{1}{1-\cos x}\,dx = \int \frac{1+\cos x}{\sin^2 x}\,dx = \int \csc^2 x\,dx + \int \frac{d\sin x}{\sin^2 x}$

$= -\cot x - \frac{1}{\sin x} + C.$

(5) $\int \frac{\sin 2x}{1+\sin^4 x}\,dx = \int \frac{2\sin x}{1+\sin^4 x}\,d\sin x = \int \frac{2t}{1+t^4}\,dt = \int \frac{1}{1+(t^2)^2}\,dt^2$

$= \arctan t^2 + C = \arctan(\sin^2 x) + C$

(6) $\int \frac{\sin^2 x\cos x}{1+\sin^3 x}\,dx = \int \frac{\sin^2 x}{1+\sin^3 x}\,d\sin x = \frac{1}{3}\int \frac{d(1+\sin^3 x)}{1+\sin^3 x} = \frac{1}{3}\ln(1+\sin^3 x) + C.$

(7) $\int \sin\alpha x \sin\beta x\,dx = \frac{1}{2}\int [\cos(\alpha-\beta)x - \cos(\alpha+\beta)x]\,dx$

$= \frac{\sin(\alpha-\beta)x}{2(\alpha-\beta)} - \frac{\sin(\alpha+\beta)x}{2(\alpha+\beta)} + C.$

(8) $\int \frac{\sin x\cos x}{2\cos^2 x + 3\sin^2 x}\,dx = \int \frac{\sin x\,d\sin x}{2+\sin^2 x} = \frac{1}{2}\int \frac{d(2+\sin^2 x)}{2+\sin^2 x}$

$= \frac{1}{2}\ln(2+\sin^2 x) + C.$

5. 计算下列不定积分:

(1) $\int x e^{-3x}\,dx = -\frac{1}{3}\int x\,de^{-3x} = -\frac{1}{3}x e^{-3x} + \frac{1}{3}\int e^{-3x}\,dx = -\frac{1}{3}xe^{-3x} - \frac{1}{9}e^{-3x} + C.$

(2) $\int x\arctan x\,dx = \frac{1}{2}\int \arctan x\,dx^2 = \frac{1}{2}x^2\arctan x - \frac{1}{2}\int x^2\cdot\frac{1}{1+x^2}\,dx$

$= \frac{1}{2}x^2\arctan x - \frac{1}{2}\int\left(1 - \frac{1}{1+x^2}\right)dx$

$= \frac{1}{2}x^2\arctan x - \frac{1}{2}x + \frac{1}{2}\arctan x + C$

(3) $\int \frac{\ln x}{x^2}\,dx = -\int \ln x\,d\left(\frac{1}{x}\right) = -\frac{\ln x}{x} + \int \frac{1}{x}\cdot\frac{1}{x}\,dx = -\frac{\ln x}{x} - \frac{1}{x} + C$

(4) $\int (\arcsin x)^2\,dx = x(\arcsin x)^2 - \int x\cdot 2\arcsin x\cdot\frac{1}{\sqrt{1-x^2}}\,dx$

$= x(\arcsin x)^2 + 2\int \arcsin x\,d\sqrt{1-x^2}$

$= x(\arcsin x)^2 + 2\sqrt{1-x^2}\arcsin x - 2\int \sqrt{1-x^2}\cdot\frac{1}{\sqrt{1-x^2}}\,dx$

$= x(\arcsin x)^2 + 2\sqrt{1-x^2}\arcsin x - 2x + C.$

(5) $\int x^2 \sin 2x\,dx = -\frac{1}{2}\int x^2\, d\cos 2x = -\frac{1}{2}x^2\cos 2x + \frac{1}{2}\int \cos 2x \cdot 2x\,dx$

$= -\frac{1}{2}x^2\cos 2x + \frac{1}{2}\int x\, d\sin 2x = -\frac{1}{2}x^2\cos 2x + \frac{1}{2}x\sin 2x - \frac{1}{2}\int \sin 2x\,dx$

$= -\frac{1}{2}x^2\cos 2x + \frac{1}{2}x\sin 2x + \frac{1}{4}\cos 2x + C.$

(6) $\int x e^x \sin x\,dx = \int x\sin x\, de^x = xe^x\sin x - \int e^x(\sin x + x\cos x)\,dx$

$= xe^x\sin x - \int e^x\sin x\,dx - \int xe^x\cos x\,dx$

$= xe^x\sin x - \int e^x\sin x\,dx - \int x\cos x\, de^x$

$= xe^x(\sin x - \cos x) - \int e^x\sin x\,dx + \int e^x(\cos x - x\sin x)\,dx$

$= xe^x(\sin x - \cos x) - \int e^x(\sin x - \cos x)\,dx - \int xe^x\sin x\,dx.$

而 $\int e^x(\sin x - \cos x)\,dx = e^x(\sin x - \cos x) - \int e^x(\cos x + \sin x)\,dx$

$= e^x(\sin x - \cos x) - e^x(\cos x + \sin x) + \int e^x(-\sin x + \cos x)\,dx$

$= -2e^x\cos x - \int e^x(\sin x - \cos x)\,dx,$

即: $\int e^x(\sin x - \cos x)\,dx = -e^x\cos x + C.$

$\therefore \int xe^x\sin x\,dx = \frac{1}{2}xe^x(\sin x - \cos x) + \frac{1}{2}e^x\cos x + C.$

(7) $\int \frac{\arcsin x}{\sqrt{1-x}}\,dx = -2\int \arcsin x\, d\sqrt{1-x} = -2\sqrt{1-x}\arcsin x + \int \frac{+2\sqrt{1-x}}{\sqrt{1-x^2}}\,dx$

$= -2\sqrt{1-x}\arcsin x + 2\int \frac{dx}{\sqrt{1+x}} = -2\sqrt{1-x}\arcsin x + 4\sqrt{1+x} + C.$

(8) $\int \frac{x\,dx}{\sin^2 x} = -\int x\, d\cot x = -x\cot x + \int \cot x\,dx = -x\cot x + \int \frac{d\sin x}{\sin x}$

$= -x\cot x + \ln|\sin x| + C.$

(9) $\int \frac{\ln\cos x}{\cos^2 x}\sin x\,dx = -\int \frac{\ln\cos x}{\cos^2 x}\,d\cos x = \int \ln\cos x\, d\left(\frac{1}{\cos x}\right)$

$= \frac{\ln\cos x}{\cos x} - \int \frac{1}{\cos x}\cdot\frac{1}{\cos x}\cdot(-\sin x)\,dx$

$= \frac{\ln\cos x}{\cos x} - \int \frac{1}{\cos^2 x}\,d\cos x = \frac{\ln\cos x}{\cos x} + \frac{1}{\cos x} + C.$

(10) $\int \sin(\ln x)\,dx = x\sin(\ln x) - \int x\cdot\cos(\ln x)\cdot\frac{1}{x}\,dx = x\sin(\ln x) - \int \cos(\ln x)\,dx$

$= x[\sin(\ln x) - \cos(\ln x)] - \int x\cdot\sin(\ln x)\cdot\frac{1}{x}\,dx$

$\therefore \int \sin(\ln x)\,dx = \frac{x}{2}[\sin(\ln x) - \cos(\ln x)] + C$

(11) $\int \frac{\arcsin e^x}{e^x}\,dx = \int \frac{\arcsin e^x}{(e^x)^2}\,de^x = -\int \arcsin e^x\, d\left(\frac{1}{e^x}\right)$

$= -\frac{\arcsin e^x}{e^x} + \int \frac{1}{e^x}\cdot\frac{e^x}{\sqrt{1-e^{2x}}}\,dx = -\frac{\arcsin e^x}{e^x} + \int \frac{dx}{\sqrt{1-e^{2x}}}$

令 $\sqrt{1-e^{2x}} = t$. 则 $dt = \frac{-2e^{2x}}{2\sqrt{1-e^{2x}}}\,dx$ 即 $dx = \frac{-\sqrt{1-e^{2x}}}{e^{2x}}\,dt$.

$\int \frac{\arcsin e^x}{e^x}\,dx = -\frac{\arcsin e^x}{e^x} - \int \frac{dt}{e^{2x}} = -\frac{\arcsin e^x}{e^x} - \int \frac{dt}{1-t^2}$

$= -\frac{\arcsin e^x}{e^x} + \frac{1}{2}\ln\frac{1-t}{1+t} + C = -\frac{\arcsin e^x}{e^x} + \frac{1}{2}\ln\frac{1-\sqrt{1-e^{2x}}}{1+\sqrt{1-e^{2x}}} + C$

(12) $\int \ln(x+\sqrt{1+x^2})\,dx = x\ln(x+\sqrt{1+x^2}) - \int x\cdot\frac{1}{x+\sqrt{1+x^2}}\cdot\left(1+\frac{2x}{2\sqrt{1+x^2}}\right)dx$

$= x\ln(x+\sqrt{1+x^2}) - \int \frac{x}{\sqrt{1+x^2}}\,dx$

$= x\ln(x+\sqrt{1+x^2}) - \sqrt{1+x^2} + C$

(13) $\int \frac{x\arccos x}{(1-x^2)^{3/2}}\,dx = -\frac{1}{2}\int \frac{\arccos x}{(1-x^2)^{3/2}}\,d(1-x^2) = \int \arccos x\, d\left(\frac{1}{\sqrt{1-x^2}}\right)$

$= \frac{\arccos x}{\sqrt{1-x^2}} + \int \frac{1}{\sqrt{1-x^2}}\cdot\frac{1}{\sqrt{1-x^2}}\,dx$

$= \frac{\arccos x}{\sqrt{1-x^2}} + \frac{1}{2}\int\left(\frac{1}{1+x} + \frac{1}{1-x}\right)dx$

$= \frac{\arccos x}{\sqrt{1-x^2}} + \frac{1}{2}\ln\left|\frac{1+x}{1-x}\right| + C$

(14) $\int \frac{xe^{2x}}{(1+2x)^2}\,dx = \frac{1}{2}\int\left(\frac{xe^{2x}}{(1+2x)^2}\right)d(1+2x) = -\frac{1}{2}\int xe^{2x}\,d\left(\frac{1}{1+2x}\right)$

$= -\frac{xe^{2x}}{2(1+2x)} + \frac{1}{2}\int \frac{1}{1+2x}\cdot(e^{2x} + 2xe^{2x})\,dx$

$= -\frac{xe^{2x}}{2(1+2x)} + \frac{1}{4}e^{2x} + C = \frac{e^{2x}}{4(1+2x)} + C$

b. 计算下列不定积分：

(1) $\int \frac{dx}{(x-2)(x+5)} = \frac{1}{7}\int \left(\frac{1}{x-2} - \frac{1}{x+5}\right)dx = \frac{1}{7}\ln\left|\frac{x-2}{x+5}\right| + C.$

(2) $\int \frac{x}{x^3-1}dx = \frac{1}{3}\int \left(\frac{1}{x-1} - \frac{x-1}{x^2+x+1}\right)dx = \frac{1}{3}\ln|x-1| - \frac{1}{6}\int \frac{2x+1-3}{x^2+x+1}dx$

$= \frac{1}{3}\ln|x-1| - \frac{1}{6}\ln|x^2+x+1| + \frac{1}{2}\int \frac{d(x+\frac{1}{2})}{(x+\frac{1}{2})^2+\frac{3}{4}}$

$= \frac{1}{6}\ln\frac{x^2-2x+1}{x^2+x+1} + \frac{\sqrt{3}}{3}\arctan\frac{2x+1}{\sqrt{3}} + C.$

(3) $\int \frac{x^4dx}{x^4+5x^2+4} = \int \left[1 - \frac{5x^2+4}{(x^2+1)(x^2+4)}\right]dx = x + \frac{1}{3}\int \left(\frac{1}{x^2+1} - \frac{16}{x^2+4}\right)dx$

$= x + \frac{1}{3}\arctan x - \frac{8}{3}\arctan\frac{x}{2} + C.$

(4) $\int \frac{x^3+4x^2-4x-1}{(x^2+1)^2}dx = \int\left\{\frac{x+4}{x^2+1} - 5\frac{x+1}{(x^2+1)^2}\right\}dx$

$= \int \frac{1}{2}\frac{d(x^2+1)}{x^2+1} + \int \frac{4dx}{1+x^2} - \frac{5}{2}\int \frac{d(x^2+1)}{(x^2+1)^2} - 5\int \frac{1}{(x^2+1)^2}dx.$

$= \frac{1}{2}\ln(1+x^2) + 4\arctan x + \frac{5}{2}\frac{1}{1+x^2} - 5\int \frac{1}{(x^2+1)^2}dx.$

令 $x = \tan\theta$. $x^2+1 = \sec^2\theta$. $dx = \sec^2\theta\,d\theta$

$\int \frac{1}{(x^2+1)^2}dx = \int \frac{\sec^2\theta\,d\theta}{\sec^4\theta} = \int \cos^2\theta\,d\theta = \frac{1}{2}\int(1+\cos 2\theta)d\theta$

$= \frac{1}{2}\theta + \frac{1}{4}\sin 2\theta + C = \frac{1}{2}\arctan x + \frac{x}{2(1+x^2)} + C$

$\therefore \int \frac{x^3+4x^2-4x-1}{(x^2+1)^2}dx = \frac{1}{2}\ln(1+x^2) + 4\arctan x + \frac{5}{2(1+x^2)} - \frac{5}{2}\arctan x - \frac{5x}{2(1+x^2)^2} + C$

$= \frac{1}{2}\ln(1+x^2) + \frac{3}{2}\arctan x + \frac{5(1-x)}{2(1+x^2)} + C.$

(5) $\int \frac{x^5dx}{x-1} = \int \left(\frac{x^5-1}{x-1} + \frac{1}{x-1}\right)dx = \int \left(x^4+x^3+x^2+1+\frac{1}{x-1}\right)dx$

$= \frac{1}{5}x^5 + \frac{1}{4}x^4 + \frac{1}{3}x^3 + \frac{1}{2}x^2 + x + \ln|x-1| + C.$

(6) $\int \frac{dx}{(1+x^2)^3}$. 令 $x = \tan\theta$. $dx = \sec^2\theta\,d\theta$. $1+x^2 = \sec^2\theta$.

$= \int \frac{\sec^2\theta\,d\theta}{\sec^6\theta} = \int \cos^4\theta\,d\theta = \frac{3}{8}\theta + \frac{1}{4}\sin 2\theta + \frac{1}{32}\sin 4\theta + C$ (利用 4(1))

$$= \frac{3}{8}\arctan x + \frac{1}{4}\cdot\frac{2x}{1+x^2} + \frac{1}{16}\cdot\frac{2x}{1+x^2}\cdot\frac{1-x^2}{1+x^2} + C$$

$$= \frac{3}{8}\arctan x + \frac{3x}{8(1+x^2)} + \frac{x}{4(1+x^2)^2} + C.$$

(7) $$\int \frac{x\,dx}{(4x^2-4x+5)^2} = \frac{1}{8}\int \frac{d(4x^2-4x+5)}{(4x^2-4x+5)^2} + \frac{1}{2}\int \frac{dx}{[(2x-1)^2+4]^2}$$

$$= -\frac{1}{8}\cdot\frac{1}{4x^2-4x+5} + \frac{1}{4}\cdot I$$

$\left(\int \frac{dx}{[(2x-1)^2+4]^2} \overset{t=2x-1}{=} \frac{1}{2}\int \frac{dt}{(t^2+4)^2} = \frac{I}{2}\right)$

$$I = \int \frac{dt}{(t^2+4)^2} = \frac{1}{4}\int \frac{t^2+4-t^2}{(t^2+4)^2}dt = \frac{1}{4}\int \frac{dt}{t^2+4} - \frac{1}{4}\int \frac{t^2}{(t^2+4)^2}dt$$

$$= \frac{1}{4}\int \frac{dt}{t^2+4} + \frac{1}{8}\int t\,d\left(\frac{1}{t^2+4}\right) = \frac{1}{4}\int \frac{dt}{t^2+4} + \frac{t}{8(t^2+4)} - \frac{1}{8}\int \frac{dt}{t^2+4}.$$

$$= \frac{t}{8(t^2+4)} + \frac{1}{8}\int \frac{dt}{t^2+4}.$$

$$\therefore \int \frac{x\,dx}{(4x^2-4x+5)^2} = -\frac{1}{8(4x^2-4x+5)} + \frac{2x-1}{32(4x^2-4x+5)} + \frac{1}{64}\arctan\left(x-\frac{1}{2}\right) + C$$

$$= \frac{2x-5}{32(4x^2-4x+5)} + \frac{1}{64}\arctan\left(x-\frac{1}{2}\right) + C.$$

(8) $$\int \frac{x^7\,dx}{(x^4+2)^3} = \frac{1}{4}\int \frac{x^4}{(x^4+2)^3}dx^4 \overset{t=x^4}{=\!=} \frac{1}{4}\int \frac{t}{(t+2)^3}dt$$

$$= \frac{1}{4}\int\left[\frac{1}{(t+2)^2} - \frac{2}{(t+2)^3}\right]dt = \frac{1}{4}\left[-\frac{1}{t+2} + \frac{1}{(t+2)^2}\right] + C$$

$$= \frac{1}{4(x^4+2)^2} - \frac{1}{4(x^4+2)} + C.$$

7. 计算下列不定积分.

(1) $$\int \frac{dx}{1+\sin^2 x} = \int \frac{2\,dx}{3-\cos 2x}.$$ 令 $t = \tan x$. $\cos 2x = \frac{1-t^2}{1+t^2}$. $dx = \frac{dt}{1+t^2}$

$$= \int \frac{2\,dt}{3+3t^2-1+t^2} = \int \frac{dt}{1+2t^2} = \frac{1}{\sqrt{2}}\arctan(\sqrt{2}\,t) + C$$

$$= \frac{\sqrt{2}}{2}\arctan(\sqrt{2}\tan x) + C.$$

(2) $\int \frac{dx}{\sin x \cos^4 x} = \int \csc x \cdot \sec^2 x \, d\tan x = \int \csc x \cdot (\tan^2 x + 1) \, d\tan x$

$= \int \csc x \, d\left(\frac{1}{3}\tan^3 x + \tan x\right) = \csc x \cdot \left(\frac{1}{3}\tan^3 x + \tan x\right)$

$- \int \left(\frac{1}{3}\tan^3 x + \tan x\right) \cdot (-\csc x \cdot \cot x) \, dx$

$= \frac{\tan^2 x + 3}{3\cos x} + \int \csc x \cdot \left(\frac{1}{3}\tan^2 x + 1\right) dx$

$= \frac{\tan^2 x + 3}{3\cos x} + \frac{1}{3}\int \csc x \cdot \sec^2 x \, dx + \frac{2}{3}\int \csc x \, dx$

$= \frac{\tan^2 x + 3}{3\cos x} + \frac{1}{3}\int \csc x \, d\tan x + \frac{2}{3}\int \csc x \, dx$

$= \frac{\tan^2 x + 3}{3\cos x} + \frac{1}{3}\csc x \cdot \tan x - \frac{1}{3}\int \tan x \cdot (-\csc x \cdot \cot x) \, dx$

$+ \frac{2}{3}\int \csc x \, dx$

$= \frac{\tan^2 x + 4}{3\cos x} + \int \csc x \, dx$

$= \frac{1}{3\cos^3 x} + \frac{1}{\cos x} + \int \frac{-d\cos x}{1 - \cos^2 x}$

$= \frac{1}{3\cos^3 x} + \frac{1}{\cos x} - \frac{1}{2}\ln\left|\frac{1+\cos x}{1-\cos x}\right| + C$

(3) $\int \frac{\sin x \, dx}{\sin x + \cos x} = \frac{1}{2}\int \left(1 + \frac{\sin x - \cos x}{\sin x + \cos x}\right) dx = \frac{x}{2} + \frac{1}{2}\int \frac{-d(\sin x + \cos x)}{\sin x + \cos x}$

$= \frac{x}{2} - \frac{1}{2}\ln|\sin x + \cos x| + C$

(4) $\int \frac{1 - \tan x}{1 + \tan x} dx = \int \frac{\cos x - \sin x}{\cos x + \sin x} dx = \int \frac{d(\sin x + \cos x)}{\sin x + \cos x} = \ln|\sin x + \cos x| + C$

(5) $\int \frac{dx}{1 + 2\cos x}$ 令 $t = \tan\frac{x}{2}$ $\cos x = \frac{1 - t^2}{1 + t^2}$ $dx = \frac{2\,dt}{1 + t^2}$

$= \int \frac{2\,dt}{1 + t^2} \cdot \frac{1}{1 + 2 \cdot \frac{1 - t^2}{1 + t^2}} = \int \frac{2\,dt}{3 - t^2} = \frac{1}{\sqrt{3}}\int \left(\frac{1}{\sqrt{3} - t} + \frac{1}{\sqrt{3} + t}\right) dt$

$= \frac{1}{\sqrt{3}}\ln\left|\frac{\sqrt{3} + t}{\sqrt{3} - t}\right| + C = \frac{1}{\sqrt{3}}\ln\left|\frac{\sqrt{3} + \tan\frac{x}{2}}{\sqrt{3} - \tan\frac{x}{2}}\right| + C$

(6) $\int \frac{dx}{(2+\cos x)\sin x} = \int \frac{\sin x\,dx}{(2+\cos x)(1-\cos^2 x)} = -\int \frac{d(\cos x)}{(2+\cos x)(1-\cos^2 x)}$ $\xleftarrow{} t=\cos x$

$= -\int \frac{dt}{(2+t)(1-t^2)} = \frac{1}{3}\int \frac{dt}{2+t} - \frac{1}{2}\int \frac{dt}{1+t} - \frac{1}{6}\int \frac{dt}{1-t}$

$= \frac{1}{3}\ln|2+t| - \frac{1}{2}\ln|1+t| + \frac{1}{6}\ln|1-t| + C$

$= \frac{1}{6}\ln\frac{(1-\cos x)(2+\cos x)^2}{(1+\cos x)^3} + C.$

(7) $\int \frac{1+\sin x}{1+\cos x}dx = \int \frac{(\cos\frac{x}{2}+\sin\frac{x}{2})^2}{2\cos^2\frac{x}{2}}dx = \frac{1}{2}\int (1+\tan\frac{x}{2})^2 dx$ $\xleftarrow{} t=\tan\frac{x}{2}$.

$= \frac{1}{2}\int (1+t)^2 \cdot \frac{2dt}{1+t^2} = \int \left(1+\frac{2t}{1+t^2}\right)dt$

$= t + \ln(1+t^2) + C = \tan\frac{x}{2} + \ln\left(1+\tan^2\frac{x}{2}\right) + C$

$= \tan\frac{x}{2} - 2\ln\cos\frac{x}{2} + C.$

(8) $\int \frac{dx}{2\sin x - \cos x + 5}$ $\xleftarrow{} t=\tan\frac{x}{2}$. $\sin x = \frac{2t}{1+t^2}$. $\cos x = \frac{1-t^2}{1+t^2}$. $dx = \frac{2dt}{1+t^2}$.

$= \int \frac{\frac{2dt}{1+t^2}}{\frac{4t}{1+t^2} - \frac{1-t^2}{1+t^2} + 5} = \int \frac{2dt}{3t^2+2t+2} = \frac{1}{3}\int \frac{dt}{(t+\frac{1}{3})^2 + \frac{5}{9}}$

$= \frac{1}{\sqrt{5}}\arctan\frac{3t+1}{\sqrt{5}} + C = \frac{1}{\sqrt{5}}\arctan\frac{3\tan\frac{x}{2}+1}{\sqrt{5}} + C.$

8. 计算下列不定积分:

(1) $\int \frac{\sqrt{x+1}-\sqrt{x-1}}{\sqrt{x+1}+\sqrt{x-1}}dx = \int \frac{1}{2}\left(\sqrt{x+1}-\sqrt{x-1}\right)^2 dx = \int \left(x - \sqrt{x^2-1}\right)dx$

$= \frac{x^2}{2} - \int \sqrt{x^2-1}\,dx$

$\xleftarrow{} I = \int \sqrt{x^2-1}\,dx = x\sqrt{x^2-1} - \int x\cdot\frac{x}{\sqrt{x^2-1}}dx = x\sqrt{x^2-1} - \int \sqrt{x^2-1}\,dx + \int \frac{dx}{\sqrt{x^2-1}}$

$= x\sqrt{x^2-1} - I + \ln\left|x+\sqrt{x^2-1}\right|$ $\quad\therefore I = \frac{x}{2}\sqrt{x^2-1} + \frac{1}{2}\ln\left|x+\sqrt{x^2-1}\right| + C.$

$\therefore \int \frac{\sqrt{x+1}-\sqrt{x-1}}{\sqrt{x+1}+\sqrt{x-1}}dx = \frac{x^2}{2} - \frac{x}{2}\sqrt{x^2-1} + \frac{1}{2}\ln\left|x+\sqrt{x^2-1}\right| + C$

(12) $\int \frac{dx}{\sqrt[3]{(x+1)^2(x-1)^4}} = \int \frac{dx}{(x^2-1)\sqrt[3]{\frac{x-1}{x+1}}}$ 令 $t=\sqrt[3]{\frac{x-1}{x+1}} \Rightarrow x=\frac{1+t^3}{1-t^3}$.

$dx=\frac{6t^2}{(1-t^3)^2}dt$

$= \int \frac{6t^2}{(1-t^3)^2}dt \cdot \frac{1}{\left[\left(\frac{1+t^3}{1-t^3}\right)^2-1\right]\cdot t} = \int \frac{6t\,dt}{(1+t^3)^2-(1-t^3)^2} = \int \frac{6t\,dt}{4t^3}$

$= \frac{3}{2}\int \frac{dt}{t^2} = -\frac{3}{2}\frac{1}{t}+C = -\frac{3}{2}\cdot\sqrt[3]{\frac{x+1}{x-1}}+C$

(13) $\int \frac{x\,dx}{\sqrt[4]{x^3(1-x)}} = \int \frac{dx}{\sqrt[4]{\frac{1-x}{x}}}$ 令 $t=\sqrt[4]{\frac{1-x}{x}} \Rightarrow x=\frac{1}{1+t^4}$.

$dx=-\frac{4t^3}{(1+t^4)^2}dt$.

$= \int \frac{1}{t}\left(-\frac{4t^3}{(1+t^4)^2}dt\right) = -4\int \frac{t^2\,dt}{(1+t^4)^2}$

$\int \frac{x\,dx}{\sqrt[4]{x^3(1-x)}} = \int \sqrt[4]{\frac{x}{1-x}}\cdot dx$ 令 $t=\sqrt[4]{\frac{x}{1-x}} \Rightarrow x=1-\frac{1}{1+t^4}=\frac{t^4}{1+t^4}$.

$dx=\frac{4t^3}{(1+t^4)^2}dt$.

$= \int t\cdot\frac{4t^3}{(1+t^4)^2}dt = -4\int t\cdot\frac{1}{(1+t^4)^2}d(1+t^4)$

$= -\int t\,d\left(\frac{1}{1+t^4}\right) = -\frac{t}{1+t^4}+\int \frac{1}{1+t^4}dt$ → (书本 P132. 例3.2.25)

$\left(= -\frac{t}{1+t^4}+\frac{\sqrt{2}}{8}\ln\left[\frac{t^2+\sqrt{2}t+1}{t^2-\sqrt{2}t+1}\right]+\frac{\sqrt{2}}{4}\arctan(\sqrt{2}t+1)+\frac{\sqrt{2}}{4}\arctan(\sqrt{2}t-1)+C\right)$

法2: $= -\frac{t}{1+t^4}+\frac{1}{2}\int \frac{t^2+1}{t^4+1}dt-\frac{1}{2}\int \frac{t^2-1}{t^4+1}dt$

$= -\frac{t}{1+t^4}+\frac{1}{2}\int \frac{1+\frac{1}{t^2}}{t^2+\frac{1}{t^2}}dx-\frac{1}{2}\int \frac{1-\frac{1}{t^2}}{t^2+\frac{1}{t^2}}dt$

$= -\frac{t}{1+t^4}+\frac{1}{2}\int \frac{d(t-\frac{1}{t})}{(t-\frac{1}{t})^2+(\sqrt{2})^2}-\frac{1}{2}\int \frac{d(t+\frac{1}{t})}{(t+\frac{1}{t})^2-(\sqrt{2})^2}$

$= -\frac{t}{1+t^4}+\frac{\sqrt{2}}{4}\arctan\frac{t-\frac{1}{t}}{\sqrt{2}}-\frac{\sqrt{2}}{8}\ln\left|\frac{t+\frac{1}{t}-\sqrt{2}}{t+\frac{1}{t}+\sqrt{2}}\right|+C$

$= -\frac{t}{1+t^4}+\frac{\sqrt{2}}{8}\ln\left[\frac{t^2+\sqrt{2}t+1}{t^2-\sqrt{2}t+1}\right]+\frac{\sqrt{2}}{4}\arctan\frac{t^2-1}{\sqrt{2}t}+C$.

将t回代即得.

(4) $\int \sqrt{4x^2+4x+5}\,dx = \int \sqrt{(2x+1)^2+4}\,dx = 2\int \sqrt{(x+\frac{1}{2})^2+1}\,dx$ $\xleftarrow{\frac{1}{2}}$ $x+\frac{1}{2}=\tan\theta$, $dx=\sec^2\theta\,d\theta$

$= 2\int \sec^3\theta\,d\theta = 2\int \frac{d\sin\theta}{(1-\sin^2\theta)^2} = \int \frac{2}{(1+\sin\theta)^2(1-\sin\theta)^2}\,d\sin\theta$

$= \frac{1}{2}\int \left[\frac{1}{1+\sin\theta} + \frac{1}{(1+\sin\theta)^2} + \frac{1}{1-\sin\theta} + \frac{1}{(1-\sin\theta)^2}\right] d\sin\theta$

$= \frac{1}{2}\left(\ln\frac{1+\sin\theta}{1-\sin\theta} - \frac{1}{1+\sin\theta} + \frac{1}{1-\sin\theta}\right) + C = \frac{1}{2}\ln\frac{1+\sin\theta}{1-\sin\theta} + \frac{\sin\theta}{\cos^2\theta} + C$

$= \frac{1}{2}\ln\left(\frac{1+\sin\theta}{\cos\theta}\right)^2 + \tan\theta\cdot\sec\theta + C = \ln|\sec\theta+\tan\theta| + \tan\theta\cdot\sec\theta + C$

$= \frac{1}{2}(x+\frac{1}{2})\sqrt{4x^2+4x+5} + \ln\left|(x+\frac{1}{2})+\sqrt{(x+\frac{1}{2})^2+1}\right| + C$

$= \frac{1}{2}(x+\frac{1}{2})\sqrt{4x^2+4x+5} + \ln\left|(2x+1)+\sqrt{4x^2+4x+5}\right| + C$

(5) $\int \frac{1-x+x^2}{\sqrt{1+x-x^2}}\,dx = \int \frac{2-(1+x-x^2)}{\sqrt{1+x-x^2}}\,dx = \int \frac{2\,dx}{\sqrt{\frac{5}{4}-(x-\frac{1}{2})^2}} - \int\sqrt{\frac{5}{4}-(x-\frac{1}{2})^2}\,dx$

$= 2\arcsin\frac{2}{\sqrt{5}}(x-\frac{1}{2}) - \frac{x-\frac{1}{2}}{2}\sqrt{\frac{5}{4}-(x-\frac{1}{2})^2} + \frac{1}{2}\cdot\frac{5}{4}\arcsin\frac{2}{\sqrt{5}}(x-\frac{1}{2}) + C$

$= \frac{11}{8}\arcsin\frac{2}{\sqrt{5}}(x-\frac{1}{2}) - \frac{x}{2}\sqrt{1+x-x^2} + \frac{1}{4}\sqrt{1+x-x^2} + C$

(6) $\int x\sqrt{x+x^2}\,dx = \int x\sqrt{(x+\frac{1}{2})^2-\frac{1}{4}}\,dx$ $\xleftarrow{\frac{1}{2}}$ $x+\frac{1}{2}=\frac{1}{2}\sec\theta$, $x=\frac{1}{2}(\sec\theta-1)$, $dx=\frac{1}{2}(\sec\theta\cdot\tan\theta)d\theta$

$= \int \frac{1}{2}(\sec\theta-1)\cdot\frac{1}{2}\tan\theta\cdot\frac{1}{2}\sec\theta\tan\theta\,d\theta$

$= \frac{1}{8}\int[\sec\theta\cdot(\sec\theta-1)\tan^2\theta]\,d\theta$

$= \frac{1}{8}\int[\sec^2\theta\tan^2\theta - \sec\theta\tan^2\theta]\,d\theta = \frac{1}{8}\int[\tan^2\theta\,d\tan\theta - (\sec^3\theta-\sec\theta)d\theta]$

$= \frac{1}{24}\tan^3\theta - \frac{1}{8}\cdot\frac{1}{2}\ln|\sec\theta+\tan\theta| - \frac{1}{8}\cdot\frac{1}{2}\tan\theta\cdot\sec\theta + \frac{1}{8}\ln|\sec\theta+\tan\theta| + C$

$= \frac{1}{24}\tan^3\theta + \frac{1}{16}\ln|\sec\theta+\tan\theta| - \frac{1}{16}\sec\theta\tan\theta$ $\quad \sec\theta = 2x+1,\ \tan\theta = 2\sqrt{x^2+x}$

$= \frac{1}{3}(x^2+x)^{\frac{3}{2}} - \frac{1}{8}(2x+1)\sqrt{x^2+x} + \frac{1}{16}\ln\left|(2x+1)+2\sqrt{x^2+x}\right| + C$

9. 计算下列不定积分:

(1) $\int \frac{\sqrt{1+\cos x}}{\sin x}\,dx = \int \frac{\sqrt{2}\cos\frac{x}{2}}{2\sin\frac{x}{2}\cos\frac{x}{2}}\,dx = \sqrt{2}\int \frac{d(\frac{x}{2})}{\sin\frac{x}{2}} = \sqrt{2}\int \csc\frac{x}{2}\,d\frac{x}{2}$

$= \sqrt{2}\ln\left|\csc\frac{x}{2} - \cot\frac{x}{2}\right| + C = \sqrt{2}\ln\left|\frac{1-\cos\frac{x}{2}}{\sin\frac{x}{2}}\right| + C$

$= \frac{\sqrt{2}}{2}\ln\frac{\sqrt{2}-\sqrt{1+\cos x}}{\sqrt{2}+\sqrt{1+\cos x}} + C.$

(2) $\int \frac{dx}{x^6(1+x^2)} = \int\left(\frac{1}{x^6} - \frac{1}{x^4} + \frac{1}{x^2} - \frac{1}{1+x^2}\right)dx = -\frac{1}{5x^5} + \frac{1}{3x^3} - \frac{1}{x} - \arctan x + C.$

(3) $\int \frac{\arctan\sqrt{x}}{\sqrt{x}(1+x)}\,dx = 2\int \frac{\arctan\sqrt{x}}{1+x}\,d\sqrt{x} = 2\int \arctan\sqrt{x}\,d\arctan\sqrt{x}$

$= (\arctan\sqrt{x})^2 + C.$

(4) $\int \frac{dx}{\sqrt{1+e^{2x}}}$ 令 $\sqrt{1+e^{2x}} = t$, $1+e^{2x} = t^2$, $e^{2x}\cdot 2dx = 2t\,dt$, $dx = \frac{t\,dt}{e^{2x}}$

$e^{2x} = t^2 - 1$, $= \frac{t\,dt}{t^2-1}$

$= \int \frac{1}{t}\cdot\frac{t\,dt}{t^2-1} = \int \frac{dt}{t^2-1} = \frac{1}{2}\int\left(\frac{1}{t-1} - \frac{1}{t+1}\right)dt$

$= \frac{1}{2}\ln\left|\frac{t-1}{t+1}\right| + C = \frac{1}{2}\ln\frac{\sqrt{1+e^{2x}}-1}{\sqrt{1+e^{2x}}+1} + C.$

(5) $\int \frac{x\ln x}{(1+x^2)^2}\,dx = \frac{1}{2}\int \frac{\ln x}{(1+x^2)^2}\,d(x^2+1) = -\frac{1}{2}\int \ln x\,d\left(\frac{1}{1+x^2}\right)$

$= -\frac{1}{2}\cdot\frac{\ln x}{1+x^2} + \frac{1}{4}\int \frac{1}{1+x^2}\cdot\frac{1}{x^2}\,dx^2 = -\frac{1}{2}\cdot\frac{\ln x}{1+x^2} + \frac{1}{4}\int\left(\frac{1}{x^2} - \frac{1}{1+x^2}\right)dx^2$

$= \frac{1}{2}\ln x + \frac{1}{2}\ln(1+x^2) + \frac{1}{4}\ln x^2 - \frac{1}{4}\ln(1+x^2)$

$= -\frac{1}{2}\cdot\frac{\ln x}{1+x^2} + \frac{1}{4}\ln\frac{x^2}{1+x^2} = -\frac{1}{4}\ln(1+x^2) + \frac{x^2\ln x}{2(1+x^2)} + C.$

(6) $\int \frac{\ln x}{(1+x^2)^{3/2}}\,dx = \int \ln x\,d\left(\frac{x}{\sqrt{1+x^2}}\right) = \frac{x\ln x}{\sqrt{1+x^2}} - \int \frac{x}{\sqrt{1+x^2}}\cdot\frac{1}{x}\,dx$

$= \frac{x\ln x}{\sqrt{1+x^2}} - \int \frac{1}{\sqrt{1+x^2}}\,dx$

$= \frac{x\ln x}{\sqrt{1+x^2}} - \ln\left(x+\sqrt{1+x^2}\right) + C.$

(7) $\int \frac{\sin^2 x}{\cos^6 x}dx = \int\left(\frac{1}{\cos^6 x} - \frac{1}{\cos^4 x}\right)dx = \int(\sec^6 x - \sec^4 x)dx$

$= \int(\sec^4 x - \sec^2 x)\,d(\tan x) = \int(1+\tan^2 x)\cdot\tan^2 x\,d\tan x$

$= \int \tan^2 x\,d\tan x + \int \tan^4 x\,d\tan x$

$= \frac{1}{5}\tan^5 x + \frac{1}{3}\tan^3 x + C$

(8) $\int \frac{dx}{x\sqrt{x^4+2x^2-1}} = \int \frac{dx}{x^3\sqrt{(1+\frac{2}{x^2}-\frac{1}{x^4})}} = -\frac{1}{2}\int \frac{d(\frac{1}{x^2})}{\sqrt{1+\frac{2}{x^2}-\frac{1}{x^4}}}$

$= -\frac{1}{2}\int \frac{d(\frac{1}{x^2})}{\sqrt{2-(\frac{1}{x^2}-1)^2}} = -\frac{1}{2}\arcsin\frac{\frac{1}{x^2}-1}{\sqrt{2}} + C$

$= -\frac{1}{2}\arcsin\frac{1-x^2}{\sqrt{2}x^2} + C = \frac{1}{2}\arcsin\frac{x^2-1}{\sqrt{2}x^2} + C$.

$\left(= \frac{1}{2}\arctan\frac{x^2-1}{\sqrt{x^4+2x^2-1}} + C\right)$

10. 已知 $f(x)$ 的一个原函数为 $\frac{\sin x}{1+x\sin x}$，求 $\int f(x)f'(x)dx$

解: $f(x) = \left(\frac{\sin x}{1+x\sin x}\right)' = \frac{\cos x(1+x\sin x) - \sin x(\sin x + x\cos x)}{(1+x\sin x)^2} = \frac{\cos x - \sin^2 x}{(1+x\sin x)^2}$.

$\int f(x)f'(x)dx = \int f(x)\,d f(x) = \frac{1}{2}f^2(x) + C = \frac{1}{2}\left[\frac{\cos x - \sin^2 x}{(1+x\sin x)^2}\right]^2 + C$.

11. 设 $f'(\sin^2 x) = \cos 2x + \tan^2 x$，求 $f(x)$.

解: 设 $t = \sin^2 x$. 则: $f'(t) = 1 - 2\sin^2 x + \frac{\sin^2 x}{1-\sin^2 x} = 1 - 2t + \frac{t}{1-t} = \frac{1}{1-t} - 2t$

$f(x) = \int f'(x)dx = \int\left(\frac{1}{1-x} - 2x\right)dx = -\ln|1-x| - x^2 + C$.

12. 设 $f(\ln x) = \frac{\ln(1+x)}{x}$，求 $\int f(x)dx$

解: 令 $t = \ln x$. 则 $x = e^t$. $f(t) = \frac{\ln(1+e^t)}{e^t}$

$\int f(x)dx = \int \frac{\ln(1+e^x)}{e^x}dx = -\int \ln(1+e^x)\,d(e^{-x})$

$$= -\frac{\ln(1+e^x)}{e^x} + \int e^{-x}\, d\left(\ln(1+e^x)\right)$$

$$= -\frac{\ln(1+e^x)}{e^x} + \int e^{-x}\cdot\frac{e^x}{1+e^x}dx = -\frac{\ln(1+e^x)}{e^x} - \int \frac{1}{e^{-x}+1}\, d(e^{-x}+1)$$

$$= -\frac{\ln(1+e^x)}{e^x} - \ln(e^{-x}+1) + C = -\ln(1+e^x)\cdot e^{-x} - \ln(1+e^x) + x + C$$

$$= -(e^{-x}+1)\ln(1+e^x) + x + C.$$

§3. 定积分的计算

习题 P.149

1. 计算下列定积分：

(1) $\int_0^{\frac{\pi}{2}} x\sin x\,dx = -\int_0^{\frac{\pi}{2}} x\,d\cos x = -x\cos x\Big|_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \cos x\,dx = \sin x\Big|_0^{\frac{\pi}{2}} = 1.$

(2) $\int_0^{\pi} (x\sin x)^2 dx = \int_0^{\pi} x^2\cdot\frac{1}{2}(1-\cos 2x)dx = \frac{1}{6}x^3\Big|_0^{\pi} - \frac{1}{4}\int_0^{\pi} x^2\, d\sin 2x$

$$= \frac{\pi^3}{6} - \frac{1}{4}x^2\sin 2x\Big|_0^{\pi} + \frac{1}{4}\int_0^{\pi}\sin 2x\cdot 2x\,dx$$

$$= \frac{\pi^3}{6} - \frac{1}{4}\int_0^{\pi} x\,d\cos 2x = \frac{\pi^3}{6} - \frac{1}{4}x\cos 2x\Big|_0^{\pi} + \frac{1}{4}\int_0^{\pi}\cos 2x\,dx$$

$$= \frac{\pi^3}{6} - \frac{\pi}{4}.$$

(3) $\int_0^{\frac{\pi}{2}} e^x\cos x\,dx = \int_0^{\frac{\pi}{2}} \cos x\,de^x = \cos x\cdot e^x\Big|_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} e^x\sin x\,dx$

$$= -1 + \int_0^{\frac{\pi}{2}} \sin x\,de^x = -1 + \sin x\cdot e^x\Big|_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} e^x\cos x\,dx.$$

$$= -1 + e^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} e^x\cos x\,dx.$$

$$\therefore \int_0^{\frac{\pi}{2}} e^x\cos x\,dx = \frac{1}{2}\left(e^{\frac{\pi}{2}} - 1\right).$$

(4) $\int_0^1 (\arcsin x)^2 dx = (x\arcsin x)^2\Big|_0^1 - \int_0^1 x\cdot 2\arcsin x\cdot\frac{1}{\sqrt{1-x^2}}dx$

$$= \frac{\pi^2}{4} + 2\int_0^1 \arcsin x\, d\sqrt{1-x^2} = \frac{\pi^2}{4} + 2\sqrt{1-x^2}\cdot\arcsin x\Big|_0^1 - 2\int_0^1 \frac{\sqrt{1-x^2}}{\sqrt{1-x^2}}dx$$

$$= \frac{\pi^2}{4} - 2x\Big|_0^1 = \frac{\pi^2}{4} - 2.$$

(5) $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{x}{\sin^2 x}\,dx = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} x\csc^2 x\,dx = -\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} x\,d\cot x = -x\cot x\Big|_{\frac{\pi}{4}}^{\frac{\pi}{3}} + \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\cos x}{\sin x}\,dx$

$= -\frac{\pi}{3}\cdot\frac{1}{\sqrt{3}} + \frac{\pi}{4}\cdot 1 + \ln|\sin x|\Big|_{\frac{\pi}{4}}^{\frac{\pi}{3}}$

$= -\frac{\sqrt{3}}{9}\pi + \frac{1}{4}\pi + \ln\frac{\frac{\sqrt{3}}{2}}{\frac{\sqrt{2}}{2}} = -\frac{\sqrt{3}}{9}\pi + \frac{\pi}{4} + \frac{1}{2}\ln\frac{3}{2}$.

(6) $\int_{-2}^{0} (x^2+2)e^{\frac{x}{2}}\,dx = 2\int_{-2}^{0} (x^2+2)\,d e^{\frac{x}{2}} = 2(x^2+2)e^{\frac{x}{2}}\Big|_{-2}^{0} - 2\int e^{\frac{x}{2}}\cdot 2x\,dx$

$= 4 - \frac{12}{e} - 8\int x\,d e^{\frac{x}{2}} = 4 - \frac{12}{e} - 8xe^{\frac{x}{2}}\Big|_{-2}^{0} + 8\int e^{\frac{x}{2}}dx$

$= 4 - \frac{12}{e} - 16e^{-1} + 8\cdot 2e^{\frac{x}{2}}\Big|_{-2}^{0} = 4 - \frac{28}{e} + 16 - \frac{16}{e} = 20 - \frac{44}{e}$.

(7) $\int_1^e \frac{1+\ln x}{x}\,dx = \int_1^e (1+\ln x)\,d\ln x = \left[\ln x + \frac{1}{2}(\ln x)^2\right]\Big|_1^e = \frac{3}{2}$.

(8) $\int_1^{e^2} \frac{\ln x}{\sqrt{x}}\,dx = 2\int_1^{e^2} \ln x\,d\sqrt{x} = 2\sqrt{x}\ln x\Big|_1^{e^2} - 2\int_1^{e^2}\frac{\sqrt{x}}{x}\,dx$

$= 2e^2 - 2\int_1^{e^2}\frac{dx}{\sqrt{x}} = 4e - 4\sqrt{x}\Big|_1^{e^2} = 4$.

2. 计算下列定积分：

(1) $\int_0^1 \frac{\sqrt{x}}{1+\sqrt{x}}\,dx$. 令 $\sqrt{x}=t$. $x=t^2$. $dx=2t\,dt$. $x: 0\mapsto 1$, $t: 0\mapsto 1$.

$= \int_0^1 \frac{t}{1+t}\cdot 2t\,dt = 2\int_0^1 \frac{t^2-1+1}{1+t}\,dt = 2\int_0^1\left[(t-1)+\frac{1}{1+t}\right]dt$

$= 2\left(\frac{1}{2}t^2 - t + \ln|1+t|\right)\Big|_0^1 = 2\ln 2 - 1$.

(2) $\int_1^{\sqrt{3}} \frac{dx}{x\sqrt{x^2+1}} = \int_1^{\sqrt{3}} \frac{dx}{x^2\sqrt{1+\frac{1}{x^2}}} = -\int_1^{\sqrt{3}} \frac{1}{\sqrt{1+\frac{1}{x^2}}}\,d\left(\frac{1}{x}\right) = -\ln\left(\frac{1}{x}+\sqrt{1+\frac{1}{x^2}}\right)\Big|_1^{\sqrt{3}} = \ln\frac{\sqrt{2}+1}{\sqrt{3}}$.

(3) $\int_0^1 \frac{x^4\,dx}{(2-x^2)^{3/2}} = +\int_0^1 x^3\,d\left(\frac{1}{\sqrt{2-x^2}}\right) = \frac{x^3}{\sqrt{2-x^2}}\Big|_0^1 - \int_0^1 \frac{3x^2}{\sqrt{2-x^2}}\,dx = 1 - \int_0^1 \frac{6}{\sqrt{2-x^2}}\,dx + 3\int_0^1 \sqrt{2-x^2}\,dx$

$= 1 - 6\arcsin\frac{x}{\sqrt{2}}\Big|_0^1 + 3\left(\frac{x}{2}\sqrt{2-x^2} + \arcsin\frac{x}{\sqrt{2}}\right)\Big|_0^1$

$= 1 - \frac{3}{2}\pi + 3\left(\frac{1}{2}+\frac{\pi}{4}\right) = \frac{5}{2} - \frac{3}{4}\pi$.

(4) $\int_0^{\frac{1}{\sqrt{2}}} \frac{\arcsin x}{(1-x^2)^{3/2}}\,dx = \int_0^{\frac{1}{\sqrt{2}}} \arcsin x\, d\left(\frac{x}{\sqrt{1-x^2}}\right) = \frac{x}{\sqrt{1-x^2}}\arcsin x\Big|_0^{\frac{1}{\sqrt{2}}} - \int_0^{\frac{1}{\sqrt{2}}} \frac{x}{\sqrt{1-x^2}}\cdot\frac{1}{\sqrt{1-x^2}}\,dx$

$= \frac{\pi}{4} - \int_0^{\frac{1}{\sqrt{2}}} \frac{x}{1-x^2}\,dx = \frac{\pi}{4} + \frac{1}{2}\int_0^{\frac{1}{\sqrt{2}}} \frac{1}{1-x^2}\,d(1-x^2)$

$= \frac{\pi}{4} + \frac{1}{2}\ln|1-x^2|\Big|_0^{\frac{1}{\sqrt{2}}} = \frac{\pi}{4} - \frac{1}{2}\ln 2.$

(5) $\int_a^{2a} \frac{\sqrt{x^2-a^2}}{x^4}\,dx = \int_a^{2a} \frac{1}{2a^2}\sqrt{1-\frac{a^2}{x^2}}\,d\left(1-\frac{a^2}{x^2}\right) = \frac{1}{3a^2}\left(1-\frac{a^2}{x^2}\right)^{3/2}\Big|_a^{2a} = \frac{\sqrt{3}}{8a^2}.$

(6) $\int_0^{\frac{\pi}{2}} \frac{dx}{2+\sin x}$ 令 $t=\tan\frac{x}{2}$. $\sin x = \frac{2t}{1+t^2}$. $dx = \frac{2\,dt}{1+t^2}$ $x: 0\mapsto\frac{\pi}{2}$, $t: 0\mapsto 1$

$= \int_0^1 \frac{2\,dt}{1+t^2}\cdot\frac{1}{2+\frac{2t}{1+t^2}} = \int_0^1 \frac{dt}{t^2+t+1} = \int_0^1 \frac{dt}{\frac{3}{4}+(t+\frac{1}{2})^2}$

$= \frac{2}{\sqrt{3}}\arctan\frac{2}{\sqrt{3}}(t+\frac{1}{2})\Big|_0^1 = \frac{2}{\sqrt{3}}\left(\arctan\sqrt{3}-\arctan\frac{\sqrt{3}}{3}\right) = \frac{\sqrt{3}}{9}\pi.$

(7) $\int_0^{\frac{\pi}{4}} \frac{\sin^2\theta\cos^2\theta}{(\cos^3\theta+\sin^3\theta)^2}\,d\theta = \int_0^{\frac{\pi}{4}} \frac{\tan^2\theta\cdot\sec^2\theta}{(1+\tan^3\theta)^2}\,d\theta$ 令 $\tan\theta=t$. $\theta: 0\mapsto\frac{\pi}{4}$, $t: 0\mapsto 1$.

$= \int_0^{\frac{\pi}{4}} \frac{\tan^2\theta}{(1+\tan^3\theta)^2}\,d\tan\theta = \int_0^1 \frac{t^2}{(1+t^3)^2}\,dt = \frac{1}{3}\int_0^1 \frac{d(1+t^3)}{(1+t^3)^2}$

$= -\frac{1}{3}\cdot\frac{1}{1+t^3}\Big|_0^1 = -\frac{1}{6}+\frac{1}{3} = \frac{1}{6}.$

(8) $\int_0^1 x^5\sqrt{1-x^2}\,dx$ 令 $x=\sin\theta$. $dx=\cos\theta\,d\theta$. $x: 0\mapsto 1$, $\theta: 0\mapsto\frac{\pi}{2}$.

$= \int_0^{\frac{\pi}{2}} \sin^5\theta\cdot\cos\theta\cdot\cos\theta\,d\theta = \int_0^{\frac{\pi}{2}} \sin^5\theta(1-\sin^2\theta)\,d\theta$

$= \int_0^{\frac{\pi}{2}} \sin^5\theta\,d\theta - \int_0^{\frac{\pi}{2}} \sin^7\theta\,d\theta = \frac{4\times 2}{5\times 3} - \frac{6\times 4\times 2}{7\times 5\times 3} = \frac{8}{105}.$

3. 计算下列定积分:

(1) $\int_{-2}^{2} |x^2-1|\,dx = 2\int_0^2 |x^2-1|\,dx = 2\int_0^1 (1-x^2)\,dx + 2\int_1^2 (x^2-1)\,dx$

$= 2\left(x-\frac{1}{3}x^3\right)\Big|_0^1 + 2\left(\frac{1}{3}x^3-x\right)\Big|_1^2 = \frac{4}{3}+\frac{4}{3}+\frac{4}{3} = 4.$

(2) $\int_{e^{-1}}^{e} |\ln x|\,dx = -\int_{e^{-1}}^1 \ln x\,dx + \int_1^e \ln x\,dx = -x\ln x\Big|_{e^{-1}}^1 + \int_{e^{-1}}^1 dx$

$+ x\ln x\Big|_1^e - \int_1^e dx = +e^{-1}\cdot(-1) + (1-e^{-1}) + e - (e-1)$

$= 2-\frac{2}{e}.$

(3) $\int_1^{16} \arctan\sqrt{\sqrt{x}-1}\,dx$ 令 $\sqrt{x}=t$. $x=t^2$. $dx=2t\,dt$ $x:16\mapsto 1$. $t:4\mapsto 1$

$$= \int_1^4 \arctan\sqrt{t-1}\cdot 2t\,dt = \int_1^4 \arctan\sqrt{t-1}\,dt^2$$

$$= t^2\arctan\sqrt{t-1}\Big|_1^4 - \int_1^4 t^2\cdot\frac{1}{1+t-1}\cdot\frac{1}{2\sqrt{t-1}}\,dt = \frac{16}{3}\pi - \int_1^4 \frac{t}{2\sqrt{t-1}}\,dt$$

$$= \frac{16}{3}\pi - \int_1^4 (t-1+1)\,d\sqrt{t-1} = \frac{16}{3}\pi - \left[\frac{1}{3}(t-1)^{3/2} + (t-1)^{\frac{1}{2}}\right]\Big|_1^4$$

$$= \frac{16}{3}\pi - 2\sqrt{3}.$$

(4) $\int_0^{\pi} 2^4\sin^2\frac{x}{2}\cos^6\frac{x}{2}\,dx = 2^5\int_0^{\pi}\left(1-\cos^2\frac{x}{2}\right)\cos^6\frac{x}{2}\,d\frac{x}{2}$ 令 $t=\frac{x}{2}$. $x:0\mapsto\pi$ $t:0\mapsto\frac{\pi}{2}$

$$= 2^5\int_0^{\frac{\pi}{2}}\cos^6 t\,dt - 2^5\int_0^{\frac{\pi}{2}}\cos^8 t\,dt = 2^5\left(\frac{5\times3\times1}{6\times4\times2}\cdot\frac{\pi}{2} - \frac{7\times5\times3\times1}{8\times6\times4\times2}\cdot\frac{\pi}{2}\right)$$

$$= 2^5\cdot\frac{15\pi}{8\times6\times8} = \frac{5}{8}\pi.$$

(5) $\int_0^1 \ln(1+\sqrt{x})\,dx = x\ln(1+\sqrt{x})\Big|_0^1 - \int_0^1 x\cdot\frac{\frac{1}{2\sqrt{x}}}{1+\sqrt{x}}\,dx = \ln 2 - \frac{1}{2}\int_0^1\frac{\sqrt{x}}{1+\sqrt{x}}\,dx$

$$= \ln 2 - \frac{1}{2}\int_0^1\left(1-\frac{1}{1+\sqrt{x}}\right)dx = \ln 2 - \frac{1}{2} + \frac{1}{2}\int_0^1\frac{2\sqrt{x}}{1+\sqrt{x}}\,d\sqrt{x}$$

$$= \ln 2 - \frac{1}{2} + \int_0^1\left(1-\frac{1}{1+\sqrt{x}}\right)d\sqrt{x} = \ln 2 - \frac{1}{2} + \left[\sqrt{x} - \ln(1+\sqrt{x})\right]\Big|_0^1$$

$$= \ln 2 - \frac{1}{2} + 1 - \ln 2 = \frac{1}{2}.$$

(6) $\int_0^{\pi}\sin^2 x\cos nx\,dx \quad (n\in N^+)$

$$= \frac{1}{2}\int_0^{\pi}(1-\cos 2x)\cos nx\,dx = \frac{1}{2}\int_0^{\pi}\cos nx\,dx - \frac{1}{2}\int_0^{\pi}\cos 2x\cos nx\,dx$$

$$= \frac{1}{2n}\sin nx\Big|_0^{\pi} - \int_0^{\pi}\frac{1}{4}\left[\cos(2+n)x + \cos(2-n)x\right]dx$$

$$= -\frac{1}{4}\int_0^{\pi}\cos(2+n)x\,dx - \frac{1}{4}\int_0^{\pi}\cos(2-n)x\,dx$$

$$= -\frac{1}{4}\cdot\frac{1}{2+n}\sin(2+n)x\Big|_0^{\pi} - \frac{1}{4}\int_0^{\pi}\cos(2-n)x\,dx = -\frac{1}{4}\int_0^{\pi}\cos(2-n)x\,dx$$

$$= \begin{cases} -\frac{1}{4}\int_0^{\pi}1\cdot dx = -\frac{\pi}{4} & (n=2) \\ -\frac{1}{4}\cdot\frac{1}{2-n}\sin(2-n)x\Big|_0^{\pi} = 0 & (n\neq 2) \end{cases}.$$

4. 用两种方法计算定积分 $I=\int_0^1 \frac{\ln(1+x)}{1+x^2}dx$.

(1) 作变换 $x=\frac{1-t}{1+t}$. (2) 作变换 $x=\tan t$.

解: (1) 令 $x=\frac{1-t}{1+t}$. $dx=-\frac{2dt}{(1+t)^2}$. $1+x=\frac{2}{1+t}$. $1+x^2=\frac{2(1+t^2)}{(1+t)^2}$. $x:0\mapsto 1$, $t:1\mapsto 0$.

$$I=\int_0^1 \frac{\ln(1+x)}{1+x^2}dx=\int_1^0 \frac{\ln\frac{2}{1+t}}{\frac{2(1+t^2)}{(1+t)^2}}\cdot\frac{-2dt}{(1+t)^2}=\int_0^1 \frac{\ln\frac{2}{1+t}}{1+t^2}dt$$

$$=\int_0^1 \frac{\ln 2-\ln(1+t)}{1+t^2}dt=\ln 2\cdot\arctan t\Big|_0^1-\int_0^1 \frac{\ln(1+t)}{1+t^2}dt$$

$$=\frac{\pi}{4}\ln 2-\int_0^1 \frac{\ln(1+t)}{1+t^2}dt=\frac{\pi}{4}\ln 2-I.$$

$$\therefore\ I=\frac{\pi}{8}\ln 2.$$

(2) 令 $x=\tan t$. $dx=\sec^2 t\,dt$. $1+x^2=\sec^2 t$. $x:0\mapsto 1$, $t:0\mapsto\frac{\pi}{4}$.

$$I=\int_0^1 \frac{\ln(1+x)}{1+x^2}dx=\int_0^{\frac{\pi}{4}} \frac{\ln(1+\tan t)}{\sec^2 t}\cdot\sec^2 t\,dt=\int_0^{\frac{\pi}{4}}\ln(1+\tan t)\,dt$$

$$=\int_0^{\frac{\pi}{4}}\ln(\cos t+\sin t)\,dt=\int_0^{\frac{\pi}{4}}\ln\cos t\,dt$$

$$=\int_0^{\frac{\pi}{4}}\ln\left[\sqrt{2}\sin\left(t+\frac{\pi}{4}\right)\right]dt-\int_0^{\frac{\pi}{4}}\ln\cos t\,dt$$

$$=\int_0^{\frac{\pi}{4}}\ln\sqrt{2}\,dt+\int_0^{\frac{\pi}{4}}\ln\sin\left(t+\frac{\pi}{4}\right)dt-\int_0^{\frac{\pi}{4}}\ln(\cos t)\,dt$$

$$=\frac{\pi}{8}\ln 2+\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\ln\sin u\,du-\int_0^{\frac{\pi}{4}}\ln\cos t\,dt \qquad \text{令 } t=\frac{\pi}{2}-y$$

$$=\frac{\pi}{8}\ln 2+\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\ln\sin u\,du-\int_{\frac{\pi}{2}}^{\frac{\pi}{4}}\ln\sin y\,d(-y)$$

$$=\frac{\pi}{8}\ln 2+\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\ln\sin u\,du-\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\ln\sin y\,dy=\frac{\pi}{8}\ln 2.$$

5. 设 f 是连续函数, 证明:

(1) 若 f 为奇函数, 则 $\int_0^x f(t)dt$ 是偶函数;

(2) 若 f 为偶函数, 则 $\int_0^x f(t)dt$ 是奇函数.

证明：(1) f 为奇函数，即 $f(x)=-f(-x)$. 设 $F(x)=\int_0^x f(t)dt$

$$F(-x)=\int_0^{-x} f(t)dt=\int_0^x f(-t)d(-t)=\int_0^x [-f(t)]d(-t)$$

$$=\int_0^x f(t)dt=F(x). \quad \therefore \int_0^x f(t)dt \text{ 为偶函数.}$$

(2) f 为偶函数，即 $f(x)=f(-x)$. 设 $F(x)=\int_0^x f(t)dt$.

$$F(-x)=\int_0^{-x} f(t)dt=\int_0^x f(-t)d(-t)=-\int_0^x f(t)dt=-F(x).$$

$\therefore \int_0^x f(t)dt$ 为奇函数.

6. 设 f 为连续函数，证明：$\int_0^a x^3 f(x^2)dx=\frac{1}{2}\int_0^{a^2} x f(x)dx$.

证明：$\int_0^a x^3 f(x^2)dx=\frac{1}{2}\int_0^a x^2 f(x^2)d(x^2)$. 设 $t=x^2$. $x: 0\mapsto a$, $t: 0\mapsto a^2$.

$$=\frac{1}{2}\int_0^{a^2} t f(t)dt=\frac{1}{2}\int_0^{a^2} x f(x)dx.$$

7. 设 f 为连续函数，证明：$\int_0^\pi x f(\sin x)dx=\frac{\pi}{2}\int_0^\pi f(\sin x)dx$.

并利用以上结果计算 $\int_0^\pi \frac{x}{1+\cos^2 x}dx$ 和 $\int_0^\pi \frac{x\sin x}{1+\cos^2 x}dx$

证明：设 $t=\pi-x$. $x: 0\mapsto\pi$, $t: \pi\mapsto 0$.

$$\int_0^\pi x f(\sin x)dx=\int_\pi^0 (\pi-t)f(\sin t)(-dt)=\int_0^\pi (\pi-t)f(\sin t)dt$$

$$=\pi\int_0^\pi f(\sin t)dt-\int_0^\pi t f(\sin t)dt \quad (\text{将变元 } t \text{ 换成 } x)$$

$$=\pi\int_0^\pi f(\sin x)dx-\int_0^\pi x f(\sin x)dx.$$

$$\therefore \int_0^\pi x f(\sin x)dx=\frac{\pi}{2}\int_0^\pi f(\sin x)dx$$

由此可得：$\int_0^\pi \frac{x}{1+\cos^2 x}dx=\frac{\pi}{2}\int_0^\pi f(\sin x)dx=\frac{\pi}{2}\int_0^\pi \frac{1}{2-\sin^2 x}dx=\frac{\pi}{2}\int_0^\pi \frac{1}{1+\cos^2 x}dx$

$$=\frac{\pi}{2}\int_0^\pi \frac{\sec^2 x}{1+\sec^2 x}dx=\frac{\pi}{2}\int_0^\pi \frac{d\tan x}{2+\tan^2 x}=\pi\int_0^{\frac{\pi}{2}} \frac{d\tan x}{2+\tan^2 x}$$

$$=\frac{\pi}{\sqrt{2}}\cdot\frac{1}{\sqrt{2}}\arctan\left(\frac{1}{\sqrt{2}}\tan x\right)\Big|_0^{\frac{\pi}{2}}=\frac{\sqrt{2}}{4}\pi^2.$$

$$\int_0^{\pi} \frac{x\sin x}{1+\cos^2 x}dx = \int_0^{\pi} \frac{x\sin x}{2-\sin^2 x}dx = \frac{\pi}{2}\int_0^{\pi} \frac{\sin x}{2-\sin^2 x}dx = -\frac{\pi}{2}\int_0^{\pi} \frac{d\cos x}{1+\cos^2 x}$$

$$= -\frac{\pi}{2}\arctan(\cos x)\Big|_0^{\pi} = -\frac{\pi}{2}\left(-\frac{\pi}{4}-\frac{\pi}{4}\right) = \frac{\pi^2}{4}.$$

8. 设 f 为连续函数，证明：

$$\int_0^x \left(\int_0^t f(u)du\right)dt = \int_0^x f(t)(x-t)dt.$$

证明：设 $F(x) = \int_0^x f(t)(x-t)dt = x\int_0^x f(t)dt - \int_0^x t f(t)dt$

$$F'(x) = \int_0^x f(t)dt + xf(x) - xf(x) = \int_0^x f(t)dt = \int_0^x f(u)du$$

$$\therefore \int_0^x \left(\int_0^t f(u)du\right)dt = F(x) = \int_0^x f(t)(x-t)dt.$$

9. 设 $f(x) = \begin{cases} xe^{-x^2} & x\geq 0 \\ \frac{1}{1+e^x} & x<0 \end{cases}$，计算 $I = \int_1^4 f(x-2)dx$

解：$I = \int_1^4 f(x-2)dx = \int_{-1}^2 f(t)dt$ 令 $t = x-2$，$x: 1\mapsto 4$，$t: -1\mapsto 2$

$$= \int_{-1}^0 \frac{1}{1+e^x}dx + \int_0^2 xe^{-x^2}dx$$

$$= -\int_{-1}^0 \frac{d(1+e^{-x})}{1+e^{-x}} + \frac{1}{2}\int_0^2 e^{-x^2}dx^2 = \ln\frac{e+1}{2} + \frac{1}{2}(1-e^{-4})$$

10. 设函数 $f(x) = \frac{1}{2}\int_0^x (x-t)^2 g(t)dt$，其中函数 $g(x)$ 在 $(-\infty,+\infty)$ 上连续，且 $g(1)=5$，$\int_0^1 g(t)dt = 2$，证明 $f'(x) = x\int_0^x g(t)dt - \int_0^x t g(t)dt$，并计算 $f''(1)$ 和 $f'''(1)$

证明：$f(x) = \frac{1}{2}\int_0^x (x^2-2tx+t^2)g(t)dt = \frac{1}{2}x^2\int_0^x g(t)dt - x\int_0^x t g(t)dt + \frac{1}{2}\int_0^x t^2 g(t)dt$

$$f'(x) = x\int_0^x g(t)dt + \frac{1}{2}x^2 g(x) - \int_0^x t g(t)dt - x^2 g(x) + \frac{1}{2}x^2 g(x)$$

$$= x\int_0^x g(t)dt - \int_0^x t g(t)dt$$

$$f''(x) = \int_0^x g(t)dt + xg(x) - xg(x) = \int_0^x g(t)dt \qquad f'''(x) = g(x)$$

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$\therefore f''(1) = \int_0^1 g(t)dt = 2. \quad f'''(1) = g(1) = -5.$

11. 设 $(0,+\infty)$ 上的连续函数 f 满足 $f(x) = \ln x - \int_1^e f(x)dx$，求 $\int_1^e f(x)dx$.

解：设 $a = \int_1^e f(x)dx$. 则 $f(x) = \ln x - a$. 两边从 1 到 e 积分：

$$\therefore a = \int_1^e f(x)dx = \int_1^e (\ln x - a)dx = \int_1^e \ln x\,dx - a\int_1^e dx$$

$$= x\ln x\Big|_1^e - \int_1^e x\cdot\frac{1}{x}dx - a\int_1^e f(x) = e - (e-1) - a(e-1)$$

$$= 1 - a(e-1).$$

$$\therefore a = \int_1^e f(x)dx = \frac{1}{e}.$$

12. ~~求 $\int_0^{\pi} x\sin x\,dx$.~~ 设函数 f 连续，且 $\int_0^1 t f(2x-t)dt = \frac{1}{2}\arctan(x^2)$. $f(1) = 1$. 求 $\int_1^2 f(x)dx$.

解：令 $u = 2x - t$，则：

$$\int_0^1 t f(2x-t)dt = -\int_{2x}^{2x-1}(2x-u)f(u)du = 2x\int_{2x-1}^{2x} f(u)du - \int_{2x-1}^{2x} u f(u)du$$

$$= \frac{1}{2}\arctan(x^2).$$

两边求导，得：$-2\int_{2x-1}^{2x} f(u)du + 2x\left[f(2x)\cdot 2 - f(2x-1)\cdot 2\right]$

$$- 2x f(2x)\cdot 2 + (2x-1)f(2x-1)\cdot 2 = \frac{1}{2}\cdot\frac{2x}{1+x^4}$$

即：$2\int_{2x-1}^{2x} f(u)du + 4x\left[f(2x) - f(2x-1)\right] - 2\left[2x f(2x) - (2x-1)f(2x-1)\right] = \frac{x}{1+x^4}$

当 $x=1$ 时 $f(1)=1$. 代入上式得：

$$2\int_1^2 f(x)dx + 4\left[f(2) - f(1)\right] - 2\left[2f(2) - f(1)\right] = \frac{1}{2}.$$

$$\therefore \int_1^2 f(x)dx = \frac{5}{4}.$$

13. 求 $\int_0^{n\pi} x|\sin x|dx$，其中 n 为正整数.

解: $\int_{2k\pi}^{2k\pi+\pi} x|\sin x|\,dx = \int_{2k\pi}^{2k\pi+\pi} x\sin x\,dx = (\sin x - x\cos x)\Big|_{2k\pi}^{2k\pi+\pi} = (4k+1)\pi.$

$\int_{2k\pi-\pi}^{2k\pi} x|\sin x|\,dx = -\int_{2k\pi-\pi}^{2k\pi} x\sin x\,dx = -(\sin x - x\cos x)\Big|_{2k\pi-\pi}^{2k\pi} = (4k-1)\pi.$

当 $n=2m$ 时,

$$\int_0^{n\pi} x|\sin x|\,dx = \sum_{k=0}^{m-1}\left(\int_{2k\pi}^{2k\pi+\pi} x|\sin x|\,dx + \int_{2k\pi+\pi}^{2k\pi+2\pi} x|\sin x|\,dx\right)$$

$$= \sum_{k=0}^{m-1}\left((4k+1)\pi + (4k+3)\pi\right) = 4m^2\pi.$$

当 $n=2m+1$ 时,

$$\int_0^{n\pi} x|\sin x|\,dx = \sum_{k=0}^{m-1}\left(\int_{2k\pi}^{2k\pi+\pi} x|\sin x|\,dx + \int_{2k\pi+\pi}^{2k\pi+2\pi} x|\sin x|\,dx\right) + \int_{2m\pi}^{2m\pi+\pi} x|\sin x|\,dx = 4m^2\pi + (4m+1)\pi = (2m+1)^2\pi.$$

$\therefore \int_0^{n\pi} x|\sin x|\,dx = n^2\pi.$

14. 设函数 $S(x)=\int_0^x |\cos t|\,dt$, 求 $\lim\limits_{x\to+\infty}\frac{S(x)}{x}$.

解: 设 $n\pi < x \le (n+1)\pi$, $n\in N^+$. 则 $\frac{x}{n}\to\pi$ ($n\to+\infty$ 时 $x\to+\infty$).

$$\int_0^{n\pi}|\cos x|\,dx = n\int_0^{\pi}|\cos x|\,dx = 2n\int_0^{\frac{\pi}{2}}|\cos x|\,dx = 2n\int_0^{\frac{\pi}{2}}\cos x\,dx$$

$$= 2n\cdot\sin x\Big|_0^{\frac{\pi}{2}} = 2n.$$

$$0 \le \int_{n\pi}^{x}|\cos x|\,dx \le \int_{n\pi}^{x} 1\cdot dx = x - n\pi \le \pi.$$

$$\therefore \quad \frac{2n}{x} \le \frac{S(x)}{x} = \frac{1}{x}\int_0^x |\cos t|\,dt \le \frac{2n+\pi}{x}$$

而 $\lim\limits_{x\to+\infty}\frac{2n}{x} = \lim\limits_{x\to+\infty}\frac{2n+\pi}{x} = \frac{2}{\pi}$. ($n\pi < x \le (n+1)\pi$, $\frac{1}{(n+1)\pi} \le \frac{1}{x} \le \frac{1}{n\pi}$, $\frac{2n}{(n+1)\pi} \le \frac{2n}{x} \le \frac{2}{\pi}$)

$$\therefore \lim_{x\to+\infty}\frac{S(x)}{x} = \frac{2}{\pi}.$$

15. 设 $I_n = \int_0^{\frac{\pi}{4}} \tan^n x\,dx$, 其中 $n\in N^+$, 证明:

(1) $I_{n+1} \le I_n$

(2) $n\ge 2$ 时, $I_n + I_{n-2} = \frac{1}{n-1}$.

证明：i) $\forall x\in[0,\frac{\pi}{4}]$ 时，$0\le\tan x\le 1$. $\therefore \tan^{n+1}x\le\tan^n x$

$\therefore \int_0^{\frac{\pi}{4}}\tan^{n+1}x\,dx\le\int_0^{\frac{\pi}{4}}\tan^n x\,dx$. …… 即 $I_{n+1}\le I_n$.

ii) $I_n=\int_0^{\frac{\pi}{4}}\tan^n x\,dx=\int_0^{\frac{\pi}{4}}\tan^{n-2}x\cdot(\sec^2x-1)\,dx$. $(n\ge 2)$

$=\int_0^{\frac{\pi}{4}}\tan^{n-2}x\,d\tan x-\int_0^{\frac{\pi}{4}}\tan^{n-2}x\,dx=\frac{1}{n-1}\tan^{n-1}x\Big|_0^{\frac{\pi}{4}}-I_{n-2}$

$=\frac{1}{n-1}-I_{n-2}$. $\therefore I_n+I_{n-2}=\frac{1}{n-1}$.

16. 设函数 f 在 $[0,1]$ 上二阶可导，且 $f''(x)<0$，$x\in[0,1]$. 证明：

$\int_0^1 f(x^2)\,dx\le f(\frac{1}{3})$.

证明：将 $f(x)$ 在 $x=\frac{1}{3}$ 处展开成一阶 Taylor 展开：

$f(x)=f(\frac{1}{3})+f'(\frac{1}{3})(x-\frac{1}{3})+\frac{1}{2}f''(\xi)(x-\frac{1}{3})^2$. …… $0<\xi<1$.

$\because f''(x)<0$. $\therefore f(x)\le f(\frac{1}{3})+f'(\frac{1}{3})(x-\frac{1}{3})$. $x\in[0,1]$.

$\forall x\in(0,1)$：$x^2\in(0,1)$ …… $\therefore f(x^2)\le f(\frac{1}{3})+f'(\frac{1}{3})(x^2-\frac{1}{3})$.

$\therefore \int_0^1 f(x^2)\,dx\le\int_0^1\left[f(\frac{1}{3})+f'(\frac{1}{3})(x^2-\frac{1}{3})\right]dx=f(\frac{1}{3})+f'(\frac{1}{3})\int_0^1(x^2-\frac{1}{3})\,dx$

$=f(\frac{1}{3})+f'(\frac{1}{3})(\frac{1}{3}x^3-\frac{1}{3}x)\Big|_0^1=f(\frac{1}{3})$.

即 $\int_0^1 f(x^2)\,dx\le f(\frac{1}{3})$.

17. 分别以 2、4、8 等分的梯形公式和抛物线公式计算积分：$\int_0^{\frac{\pi}{2}}\frac{\sin x}{x}\,dx$.

解：梯形公式：$\int_a^b f(x)\,dx=\frac{b-a}{n}\left[\frac{y_0}{2}+(y_1+y_2+\cdots+y_{n-1})+\frac{y_n}{2}\right]$

抛物线公式：$\int_a^b f(x)\,dx=\frac{b-a}{6n}\left[y_0+4\sum_{i=1}^{n}y_{2i-1}+2\sum_{i=1}^{n-1}y_{2i}+y_{2n}\right]$

	0	1	2	3	4	5	6	7	8
x_i:	0	$\frac{\pi}{16}$	$\frac{\pi}{8}$	$\frac{3\pi}{16}$	$\frac{\pi}{4}$	$\frac{5\pi}{16}$	$\frac{3\pi}{8}$	$\frac{7\pi}{16}$	$\frac{\pi}{2}$
y_i:	1	0.9936	0.9745	0.9432	0.9003	0.8469	0.7842	0.7136	0.6366
二等分	y_0				y_1				y_2
四等分	y_0		y_1		y_2		y_3		y_4
八等分	y_0	y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8

梯形公式：2等分. $I=\frac{\frac{\pi}{2}}{2}\left[\frac{y_0}{2}+y_1+\frac{y_2}{2}\right]=1.350$ $n=2$

4等分. $I=\frac{\frac{\pi}{2}}{4}\left[\frac{y_0}{2}+y_1+y_2+y_3+\frac{y_4}{2}\right]=1.366$ $n=4$

8等分. $I=\frac{\frac{\pi}{2}}{8}\left[\frac{y_0}{2}+y_1+y_2+y_3+y_4+y_5+y_6+y_7+\frac{y_8}{2}\right]=1.369$ $n=8$

抛物线公式：2等分. $I=\frac{\frac{\pi}{2}}{6\times 1}\left[y_0+4y_1+y_2\right]=1.371$ $n=1$

4等分. $I=\frac{\frac{\pi}{2}}{6\times 2}\left[y_0+4y_1+2y_2+4y_3+y_4\right]=1.371$ $n=2$

8等分. $I=\frac{\frac{\pi}{2}}{6\times 4}\left[y_0+4y_1+2y_2+4y_3+2y_4+4y_5+2y_6+4y_7+y_8\right]=1.371$ $n=4$

§4 定积分的应用

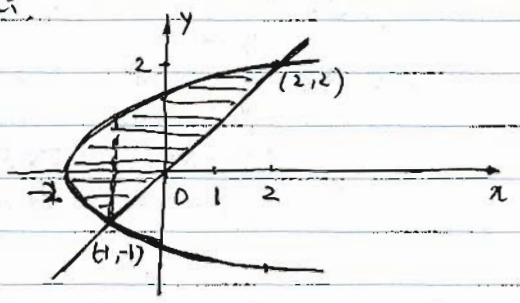
习题：P.167.

1. 求抛物线 $y^2=x+2$ 与直线 $x-y=0$ 所围图形的面积.

解：$S=\int_{-2}^{-1}2\sqrt{x+2}\,dx+\int_{-1}^{2}\left(\sqrt{x+2}-x\right)dx$

$=2\cdot\frac{2}{3}(x+2)^{\frac{3}{2}}\Big|_{-2}^{-1}+\frac{2}{3}(x+2)^{\frac{3}{2}}\Big|_{-1}^{2}-\frac{1}{2}x^2\Big|_{-1}^{2}$

$=\frac{4}{3}+\frac{14}{3}=\frac{3}{2}=\frac{9}{2}$.



2. 求抛物线 $y=-x^2-4x-3$ 与其在 $x=0$ 和 $x=3$ 处各点的两条切线围成的图形的面积.

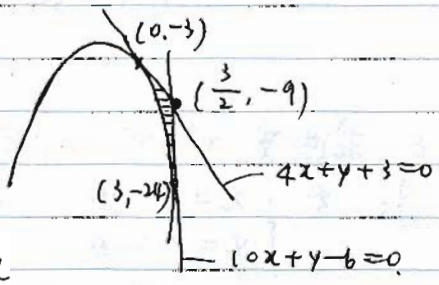
解：$y=-x^2-4x-3$. $x=0$, $y=-3$; $x=3$, $y=-24$.

$y'=-2x-4$. $y'(0)=-4$. $y'(3)=-10$.

过 $(0,-3)$ 处切线：$4x+y+3=0$

过 $(3,-24)$ 处切线：$10x+y-b=0$.

} 交点 $(\frac{3}{2},-9)$



$S=\int_0^{\frac{3}{2}}\left(-4x-3+x^2+4x+3\right)dx+\int_{\frac{3}{2}}^{3}\left(-10x+b+x^2+4x+3\right)dx$

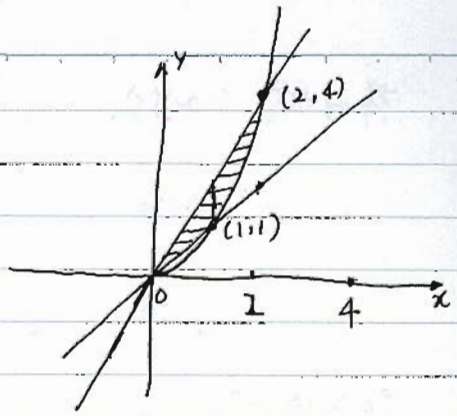
$=\int_0^{\frac{3}{2}}(x^2)\,dx+\int_{\frac{3}{2}}^{3}(x^2-6x+9)\,dx=\int_0^{\frac{3}{2}}x^2\,dx+\int_{\frac{3}{2}}^{3}(x-3)^2\,dx$

$=\frac{1}{3}x^3\Big|_0^{\frac{3}{2}}+\frac{1}{3}(x-3)^3\Big|_{\frac{3}{2}}^{3}=\frac{9}{4}$.

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3. 求曲线 $y=x^2$ 与直线 $y=x$, $y=2x$ 所围图形的面积.



解: $S=\int_0^1 (2x-x)\,dx+\int_1^2 (2x-x^2)\,dx$

$=\int_0^1 x\,dx+\int_1^2 (2x-x^2)\,dx$

$=\frac{1}{2}x^2\Big|_0^1+\left(x^2-\frac{1}{3}x^3\right)\Big|_1^2=\frac{1}{2}+\left(4-\frac{8}{3}\right)-\left(1-\frac{1}{3}\right)$

$=\frac{7}{6}$.

4. 求星形线 $x=a\cos^3 t$, $y=a\sin^3 t$ 所围图形的面积 ($a>0$).

解: $A=\frac{1}{2}\int_\alpha^\beta r^2\,d\theta=\frac{1}{2}\int_0^{2\pi}(x^2+y^2)\,d\theta=\frac{1}{2}\int_0^{2\pi}a^2(\cos^6 t+\sin^6 t)\,dt$

$=\frac{a^2}{2}\int_0^{2\pi}(\cos^6 t+\sin^6 t)\,dt=2a^2\int_0^{\frac{\pi}{2}}(\sin^6 t+\cos^6 t)\,dt=4a^2\int_0^{\frac{\pi}{2}}\cos^6 t\,dt$

$=4a^2\cdot\frac{5\times3\times1}{6\times4\times2}\cdot\frac{\pi}{2}=$

$S=\left|\int_0^{2\pi}y(t)\,d\,x(t)\right|=-\int_0^{2\pi}|y(t)\cdot x'(t)|\,dt=\int_0^{2\pi}a\sin^3 t\cdot 3a\cos^2 t(\sin t)\,dt$

$=3a^2\int_0^{2\pi}\sin^4 t(1-\sin^2 t)\,dt=3a^2\cdot 4\int_0^{\frac{\pi}{2}}(\sin^4 t-\sin^6 t)\,dt$

$=12a^2\cdot\left(\frac{3\times1}{4\times2}-\frac{5\times3\times1}{6\times4\times2}\right)\cdot\frac{\pi}{2}=\frac{3\pi a^2}{8}$.

5. 求 $r=a\sin 3\theta$ $(0\le\theta\le\frac{\pi}{3})$ 围成的图形的面积.

解: $S=\frac{1}{2}\int_0^{\frac{\pi}{3}}r^2\,d\theta=\frac{1}{2}\int_0^{\frac{\pi}{3}}a^2\sin^2 3\theta\,d\theta=\frac{a^2}{2}\int_0^{\frac{\pi}{3}}\frac{1}{2}(1-\cos 6\theta)\,d\theta$

$=\frac{a^2}{4}\left(\theta-\frac{1}{6}\sin 6\theta\right)\Big|_0^{\frac{\pi}{3}}=\frac{a^2}{4}\cdot\frac{\pi}{3}=\frac{\pi a^2}{12}$.

6. 求曲线 $x^4+y^4=a^2(x^2+y^2)$ 围成的图形的面积.

解: 令 $\begin{cases}x=r\cos\theta\\ y=r\sin\theta\end{cases}$ 则反曲线为: $r^4(\cos^4\theta+\sin^4\theta)=a^2\cdot r^2(\cos^2\theta+\sin^2\theta)$

即: $r^2=\frac{a^2}{\cos^4\theta+\sin^4\theta}$ ····· $0\le\theta\le2\pi$.

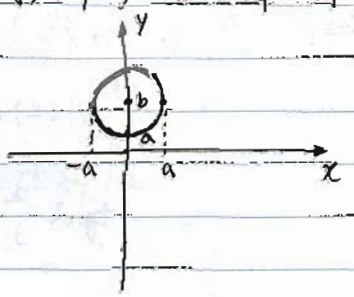
$S=\frac{1}{2}\int_0^{2\pi}r^2\,d\theta=\frac{1}{2}\int_0^{2\pi}\frac{a^2}{\cos^4\theta+\sin^4\theta}\,d\theta=2\int_0^{\frac{\pi}{2}}\frac{a^2}{\cos^4\theta+\sin^4\theta}\,d\theta$

$$= 2a^2 \int_0^{\frac{\pi}{2}} \frac{\tan^2\theta+1}{\tan^4\theta+1} d\tan\theta = 2a^2 \int_0^{\frac{\pi}{2}} \frac{d\left(\tan\theta - \frac{1}{\tan\theta}\right)}{\left(\tan\theta - \frac{1}{\tan\theta}\right)^2 + 2}$$

$$= 2a^2 \cdot \frac{1}{\sqrt{2}} \arctan \frac{1}{\sqrt{2}}\left(\tan\theta - \frac{1}{\tan\theta}\right)\Big|_0^{\frac{\pi}{2}} = \sqrt{2}a^2 \cdot \arctan\frac{\infty}{\sqrt{2}} - \sqrt{2}a^2 \cdot \arctan\left(-\frac{\infty}{\sqrt{2}}\right)$$

$$= \sqrt{2}a^2 \cdot \frac{\pi}{2} - \sqrt{2}a^2 \cdot \left(-\frac{\pi}{2}\right) = \sqrt{2}\pi a^2.$$

7. 求由圆盘 $x^2+(y-b)^2 \le a^2$ $(b>a>0)$ 绕 x 轴旋转一周所得旋转体的体积.



解: $V = \pi \int_{-a}^{a} \left[\left(b+\sqrt{a^2-x^2}\right)^2 - \left(b-\sqrt{a^2-x^2}\right)^2\right] dx$

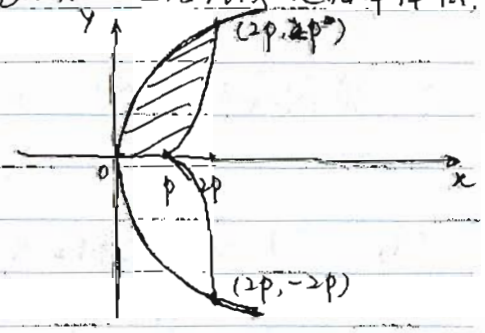
$= \pi \int_{-a}^{a} \left[2b^2 + 2(a^2-x^2)\right] dx$

$2\pi\left(a^2+b^2\right)x - \frac{1}{3}x^3\Big|_{-a}^{a}$

$$= 4\pi b \int_{-a}^{a} \sqrt{a^2-x^2}\, dx = 4\pi b\left(\frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\arcsin\frac{x}{a}\right)\Big|_{-a}^{a}$$

$$= 4\pi b \cdot \frac{a^2}{2}\left(\frac{\pi}{2}+\frac{\pi}{2}\right) = 2\pi^2 a^2 b.$$

8. 求曲线 $y^2 = 2px$ 与 $py^2 = 4(x-p)^3$ $(p>0)$ 所围图形绕 x 轴旋转所成旋转体体积.



解: $py^2 = 4(x-p)^3 \Rightarrow y = \pm\sqrt{\frac{4}{p}}\cdot(x-p)^{3/2}$.

$y' = \pm\sqrt{\frac{4}{p}}\cdot\frac{3}{2}(x-p)^{\frac{1}{2}} = 0 \Rightarrow x = p$.

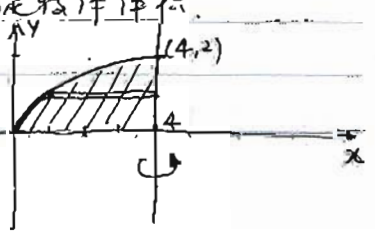
$\forall x > p,\ y' > 0$. ↗

$y'' = \pm\sqrt{\frac{4}{p}}\cdot\frac{3}{2}\cdot\frac{1}{2}(x-p)^{-\frac{1}{2}} \neq 0$.

$$V = \pi\int_0^{2p}(2px)\,dx - \pi\int_p^{2p}\frac{4}{p}(x-p)^3\,dx$$

$$= 2p\pi\cdot\frac{1}{2}x^2\Big|_0^{2p} - \frac{4\pi}{p}\cdot\frac{1}{4}(x-p)^4\Big|_p^{2p} = 4\pi p^3 - \pi p^3 = 3\pi p^3.$$

9. 求 $y=\sqrt{x}$ 与 x 轴及直线 $x=4$ 围成的图形绕 $x=4$ 旋转所得旋转体体积.



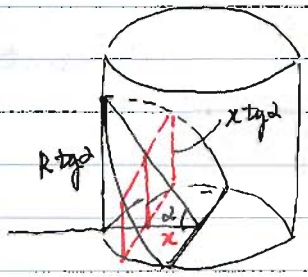
解: $V = \int_0^2 \pi(4-x)^2\,dy = \pi\int_0^2 (4-y^2)^2\,dy = \pi\int_0^2 (y^4-8y^2+16)\,dy$

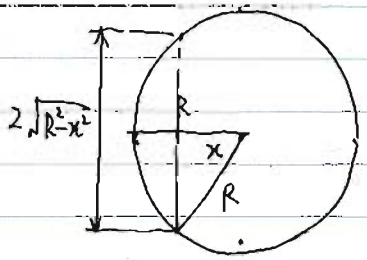
$$= \pi\left(\frac{1}{5}y^5 - \frac{8}{3}y^3 + 16y\right)\Big|_0^2 = \frac{256}{15}\pi.$$

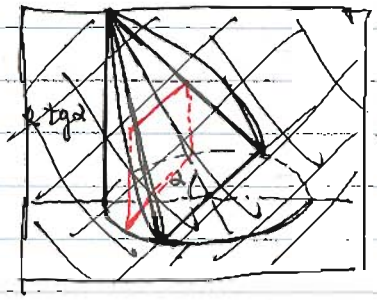
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10. 设圆柱体的半径为 R，通过底面的一条直径，作一与底面夹角为 α $(\alpha<\frac{\pi}{2})$ 的平面，从圆柱上截出一截楔体，求该楔体的体积. （已知平行截面面积求体积）$(V=\int_a^b A(x)dx)$







解：截楔体的截面为一矩形：宽：$x\tan\alpha$，长：$2\sqrt{R^2-x^2}$　　$x: 0\mapsto R$

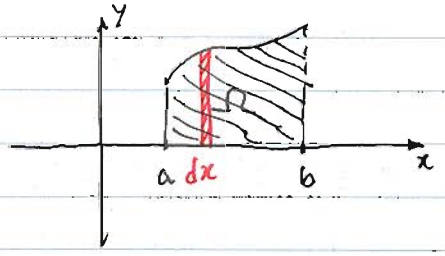
$$V=\int_0^R A(x)\,dx=\int_0^R x\tan\alpha\cdot 2\sqrt{R^2-x^2}\,dx=-\tan\alpha\int_0^R (R^2-x^2)^{\frac{1}{2}}\,d(R^2-x^2)$$

$$=-\tan\alpha\cdot\frac{2}{3}(R^2-x^2)^{\frac{3}{2}}\Big|_0^R=\frac{2}{3}R^3\tan\alpha.$$

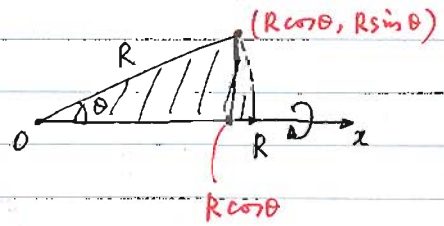
11. 设 f 为连续函数，平面区域 $\Omega=\{(x,y)\mid 0\le y\le f(x),\ a\le x\le b\}$ $(a>0)$. 将 Ω 绕 y 轴一周，则所得空间区域的体积为 $V=\int_a^b 2\pi x f(x)\,dx$.

证明：$dV=2\pi x\cdot f(x)\,dx$　（圆柱体的侧面积）.

$\therefore V=\int_a^b 2\pi x f(x)\,dx$



12. 将圆扇形 $\{(r,\theta)\mid 0\le\theta\le\alpha,\ 0\le r\le R\}$ 绕极轴旋转一周，试导出所得立体的体积为 $\frac{2}{3}\pi R^3(1-\cos\alpha)$.



解：$y=R\cos\alpha$ 为半径 $R\sin\alpha$ 圆形体积

$$V_1=\frac{1}{3}\pi(R\sin\alpha)^2\cdot R\cos\alpha=\frac{1}{3}\pi R^3\sin^2\alpha\cos\alpha.$$

圆弧绕极轴体积

$$V_2=\int_\alpha^0 \pi y_{(\theta)}^2\,dx_{(\theta)}.\qquad \theta:\alpha\mapsto 0.\qquad y(\theta)=R\sin\theta.\quad x(\theta)=R\cos\theta.$$

$$=\int_\alpha^0 \pi R^2\sin^2\theta\,d\,R\cos\theta=\pi R^3\int_\alpha^0(1-\cos^2\theta)\,d\cos\theta$$

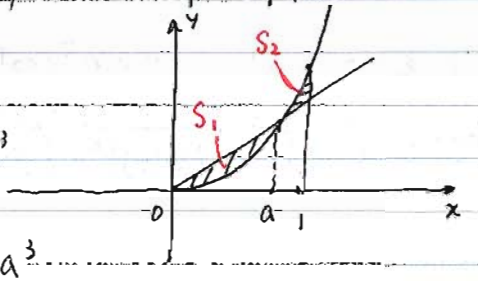
$$=\pi R^3\left(\cos\theta-\frac{1}{3}\cos^3\theta\right)\Big|_\alpha^0=\pi R^3\left(\frac{2}{3}-\cos\alpha+\frac{1}{3}\cos^3\alpha\right)$$

$$\therefore V=V_1+V_2=\pi R^3\left[\frac{1}{3}(1-\cos^2\alpha)\cos\alpha+\frac{2}{3}-\cos\alpha+\frac{1}{3}\cos^3\alpha\right]=\frac{2}{3}\pi R^3(1-\cos\alpha)$$

13. 设直线 $y=ax$ $(a<1)$ 与抛物线 $y=x^2$ 所围成的图形的面积为 S_1，且它与直线 $x=1$ 所围成图形的面积为 S_2.

(1) 试确定 a 的值，使得 S_1+S_2 达到最小，并求出最小值；

(2) 求该最小值所对应的平面图形绕 x 轴旋转一周所得旋转体的体积.



解：(1) $S_1=\int_0^a (ax-x^2)\,dx=\left(\frac{1}{2}ax^2-\frac{1}{3}x^3\right)\Big|_0^a=\frac{1}{6}a^3$.

$S_2=\int_a^1 (x^2-ax)\,dx=\left(\frac{1}{3}x^3-\frac{1}{2}ax^2\right)\Big|_a^1=\frac{1}{3}-\frac{1}{2}a+\frac{1}{6}a^3$

$S=S_1+S_2=\frac{1}{6}a^3+\frac{1}{3}-\frac{1}{2}a+\frac{1}{6}a^3=\frac{1}{3}-\frac{1}{2}a+\frac{1}{3}a^3$.

$\frac{dS}{da}=-\frac{1}{2}+a^2=0 \Rightarrow a=\pm\frac{1}{\sqrt{2}}$.

$\frac{d^2S}{da^2}=2a$. 当 $a=\frac{1}{\sqrt{2}}$ 时，$\frac{d^2S}{da^2}=\frac{2}{\sqrt{2}}=\sqrt{2}>0$. S 达到极小值.

$\therefore$ 当 $a=\frac{\sqrt{2}}{2}$ 时，$S_{min}=(S_1+S_2)_{min}=\frac{1}{3}-\frac{1}{2}a+\frac{1}{3}a^3=\frac{1}{3}-\frac{\sqrt{2}}{6}$.

(2) $V=\int_0^a \pi\left[(ax)^2-x^4\right]dx+\int_a^1 \pi\left[x^4-(ax)^2\right]dx$

$=\pi\int_0^a (a^2x^2-x^4)\,dx+\pi\int_a^1 (x^4-a^2x^2)\,dx$

$=\pi\left(\frac{a^2}{3}x^3-\frac{1}{5}x^5\right)\Big|_0^a+\pi\left(\frac{1}{5}x^5-\frac{a^2}{3}x^3\right)\Big|_a^1$

$=\pi\left(\frac{a^5}{3}-\frac{a^5}{5}\right)+\pi\left(\frac{1}{5}-\frac{a^2}{3}-\frac{a^5}{5}+\frac{a^5}{3}\right)=\pi\left(\frac{1}{5}-\frac{a^2}{3}+\frac{4}{15}a^5\right)$

$=\pi\left(\frac{1}{5}-\frac{1}{3}\cdot\frac{1}{2}+\frac{4}{15}\cdot\frac{1}{4}\cdot\frac{\sqrt{2}}{2}\right)=\pi\left(\frac{1}{30}+\frac{\sqrt{2}}{30}\right)=\frac{1+\sqrt{2}}{30}\pi$.

14. 求下列曲线段的弧长：

(1) $y=\frac{2}{3}(x-1)^{\frac{3}{2}}$. $1\le x\le 2$.

解：$s=\int_1^2 \sqrt{1+(y')^2}\,dx$. $y'=\frac{2}{3}\cdot\frac{3}{2}(x-1)^{\frac{1}{2}}=\sqrt{x-1}$.

$=\int_1^2 \sqrt{1+x-1}\,dx=\int_1^2 \sqrt{x}\,dx=\frac{2}{3}x^{\frac{3}{2}}\Big|_1^2=\frac{2}{3}(2\sqrt{2}-1)$.

(2) $y=\int_{-\frac{\pi}{2}}^{x}\sqrt{\cos t}\,dt$ $\left(-\frac{\pi}{2}\le x\le\frac{\pi}{2}\right)$.

解：$y'=\sqrt{\cos x}$. $(y')^2=\cos x$.

$$s=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\sqrt{1+(y')^2}\,dx=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\sqrt{1+\cos x}\,dx=\sqrt{2}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos\frac{x}{2}\,dx$$

$$=2\sqrt{2}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos\frac{x}{2}\,d\left(\frac{x}{2}\right)=2\sqrt{2}\sin\frac{x}{2}\Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}}=2\sqrt{2}\left(\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}\right)=4.$$

(3) $x=e^t\sin t$, $y=e^t\cos t$, $0\le t\le\pi$.

解:
$$s=\int_0^{\pi}\sqrt{(x'(t))^2+(y'(t))^2}\,dt=\int_0^{\pi}\sqrt{e^{2t}(\sin t+\cos t)^2+e^{2t}(\cos t-\sin t)^2}\,dt$$

$$=\int_0^{\pi}e^t\sqrt{2}\,dt=\sqrt{2}e^t\Big|_0^{\pi}=\sqrt{2}(e^{\pi}-1).$$

(4) 星形线 $\begin{cases}x=a\cos^3 t,\\ y=a\sin^3 t\end{cases}$ 的周长.

解:
$$s=\int_0^{2\pi}\sqrt{(x'(t))^2+(y'(t))^2}\,dt=\int_0^{2\pi}\sqrt{(3a\cos^2 t\sin t)^2+(3a\sin^2 t\cos t)^2}\,dt$$

$$=3a\int_0^{2\pi}\sqrt{\cos^4 t\sin^2 t+\sin^4 t\cos^2 t}\,dt=3a\int_0^{2\pi}|\sin t\cdot\cos t|\,dt$$

$$=12a\int_0^{\frac{\pi}{2}}\sin t\cos t\,dt=6a\int_0^{\frac{\pi}{2}}\sin 2t\,dt=3a\cdot\cos 2t\Big|_{\frac{\pi}{2}}^{0}=6a.$$

(5) $r=e^{2\theta}$, $0\le\theta\le 2\pi$.

解:
$$s=\int_0^{2\pi}\sqrt{r^2+(r')^2}\,d\theta=\int_0^{2\pi}\sqrt{e^{4\theta}+4e^{4\theta}}\,d\theta=\sqrt{5}\int_0^{2\pi}e^{2\theta}\,d\theta$$

$$=\frac{\sqrt{5}}{2}e^{2\theta}\Big|_0^{2\pi}=\frac{\sqrt{5}}{2}(e^{4\pi}-1).$$

15. 求下列曲线在指定点的曲率和曲率半径:

(1) $xy=4$. 在点 $(2,2)$.

解: $y=\frac{4}{x}$. $K=\frac{|y''|}{[1+(y')^2]^{3/2}}=\frac{\frac{8}{x^3}}{\left[1+\frac{16}{x^4}\right]^{3/2}}$. 当 $x=2$ 时: $K=\frac{1}{(1+1)^{3/2}}=\frac{\sqrt{2}}{4}$.

$R=\frac{1}{K}=2\sqrt{2}$.

(2) $x=a(t-\sin t)$

$y=a(1-\cos t)$ $(a>0)$. 在 $t=\frac{\pi}{2}$ 对应的点

解:
$$K=\frac{|x'y''-x''y'|}{(x'^2+y'^2)^{3/2}}=\frac{|a(1-\cos t)\cdot a\cos t-a\sin t\cdot a\sin t|}{[a^2(1-\cos t)^2+a^2\sin^2 t]^{3/2}}$$

$$= \frac{a^2(1-\cos t)}{\left[2a^2(1-\cos t)\right]^{3/2}} = \frac{a^2(1-\cos t)}{2a^2(1-\cos t)\cdot\sqrt{2}a\cdot\sqrt{1-\cos t}} = \frac{\sqrt{2}}{4a}\cdot\frac{1}{\sqrt{1-\cos t}}.$$

当 $t=\frac{\pi}{2}$ 时. $K = \frac{\sqrt{2}}{4a}\cdot\frac{1}{\sqrt{1-0}} = \frac{\sqrt{2}}{4a}$ $\quad R = \frac{1}{K} = 2\sqrt{2}a$.

16. 求下列曲线的曲率和曲率半径:

(1) 抛物线 $y^2 = 2px$ $(p>0)$.

解: $K = \frac{|y''|}{(1+y'^2)^{3/2}}$ $\quad 2y\cdot y' = 2p$. $\quad y' = \frac{p}{y} = \sqrt{\frac{p}{2x}} = \sqrt{\frac{p}{2}}\cdot x^{-\frac{1}{2}}$. $\quad y'' = -\frac{\sqrt{2p}}{4x\sqrt{x}}$

$$K = \frac{\sqrt{2}}{4}\cdot\frac{\sqrt{p}}{x\sqrt{x}}\cdot\left(1+\frac{p}{2x}\right)^{-\frac{3}{2}} = \frac{\sqrt{p}}{(2x+p)^{3/2}} \qquad R = \frac{1}{K} = \frac{(2x+p)^{3/2}}{\sqrt{p}}$$

(2) 双曲线 $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

解: $\frac{2x}{a^2} - \frac{2y}{b^2}\cdot y' = 0$. $\quad y' = \frac{b^2}{a^2}\cdot\frac{x}{y}$ $\quad y'' = \frac{b^2}{a^2}\cdot\frac{y-xy'}{y^2} = \frac{y-xy'}{x^2-a^2} = -\frac{b^2}{y(x^2-a^2)}$

$$K = \frac{|y''|}{(1+y'^2)^{3/2}} = \frac{b^2}{y(x^2-a^2)}\cdot\frac{1}{\left(1+\frac{b^2}{a^2}\cdot\frac{x^2}{y^2}\right)^{3/2}} = \frac{a^4b}{\left[(a^2+b^2)x^2-a^4\right]^{3/2}}$$

$$R = \frac{1}{K} = \frac{\left[(a^2+b^2)x^2-a^4\right]^{3/2}}{a^4b}$$

或: 令 $x = a\sec t$, $y = b\tan t$. 则:

$$\frac{dy}{dx} = \frac{dy}{dt}\Big/\frac{dx}{dt} = \frac{b}{a}\cdot\frac{\sec^2 t}{\sec t\cdot\tan t} = \frac{b}{a}\cdot\csc t.$$

$$\frac{d^2y}{dx^2} = \frac{d\left(\frac{dy}{dx}\right)}{dt}\Big/\frac{dx}{dt} = \frac{b}{a^2}\cdot\frac{-\csc t\cdot\cot t}{\sec t\cdot\tan t} = -\frac{b}{a^2}\cot^3 t.$$

$$K = \frac{|y''|}{(1+y'^2)^{3/2}} = \frac{\frac{b}{a^2}\cdot\cot^3 t}{\left[1+\frac{b^2}{a^2}\csc^2 t\right]^{3/2}} = \frac{ab\left|\cot t\right|^3}{(a^2+b^2\csc^2 t)^{3/2}} = \frac{a^4b}{\left[(a^2+b^2)x^2-a^4\right]^{3/2}}$$

(3) 星形线 $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ $(a>0)$.

解: 令 $x = a\cos^3 t$, $y = a\sin^3 t$.

$$\frac{dy}{dx} = \frac{3a\sin^2 t\cdot\cos t}{-3a\cos^2 t\cdot\sin t} = -\tan t. \qquad \frac{d^2y}{dx^2} = \frac{-\sec^2 t}{3a\cos^2 t\cdot(-\sin t)} = \frac{1}{3a}\cdot\frac{1}{\sin t\cos^4 t}$$

$$K=\frac{|y''|}{(1+y'^2)^{3/2}}=\frac{\frac{1}{3a}\left|\frac{1}{\sin t\cos^2 t}\right|}{(1+\tan^2 t)^{3/2}}=\frac{1}{3a}\left|\frac{1}{\sin t\cos^3 t\ \sec^3 t}\right|=\frac{1}{3a}\left|\frac{1}{\sin t\cos t}\right|$$

$$=\frac{1}{3a}\cdot\sqrt[3]{\frac{a^2}{|xy|}}=\frac{1}{3}\cdot\frac{1}{\sqrt[3]{|axy|}}.\qquad R=\frac{1}{K}=3\sqrt[3]{|axy|}.$$

(4) 求渐开线 $x=a(\cos t+t\sin t)$，$y=a(\sin t-t\cos t)$. $(a>0)$.

解: $\frac{dy}{dx}=\frac{a(\cos t-\cos t+t\sin t)}{a(-\sin t+\sin t+t\cos t)}=\tan t$. $\frac{d^2y}{dx^2}=\frac{\sec^2 t}{at\cos t}=\frac{1}{at\cos^3 t}$.

$$K=\frac{|y''|}{(1+y'^2)^{3/2}}=\frac{\left|\frac{1}{at\cos^3 t}\right|}{(1+\tan^2 t)^{3/2}}=\frac{1}{at}.\qquad R=\frac{1}{K}=at.$$

17. 求曲线 $y=\ln x$ 在点 $(1,0)$ 处的曲率圆方程.

解: $y'=\frac{1}{x}$. $y''=-\frac{1}{x^2}$. 在 $(1,0)$ 处. $K=\frac{|y''|}{(1+y'^2)^{3/2}}=\frac{1}{(1+1)^{3/2}}=\frac{\sqrt{2}}{4}$. $R=\frac{1}{K}=2\sqrt{2}$.

$y=\ln x$ 在点 $(1,0)$ 处切线斜率为 $y'(1)=1$.

法线方程为 $y=-x+1$.

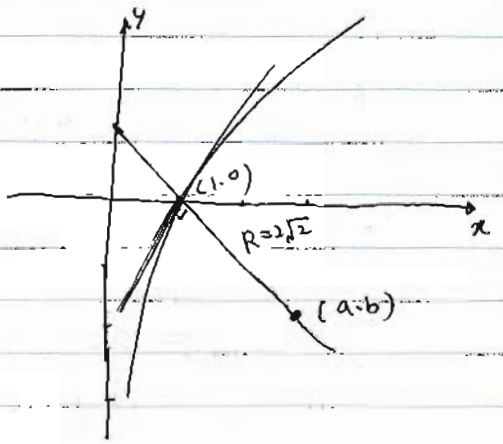
设曲率圆的圆心为 (a,b). 则 $b=-a+1$

曲率圆方程. $(x-a)^2+(y-b)^2=8$.

$(1,0)$ 为其点. $(1-a)^2+b^2=8$.

$\therefore a=3$, $b=-2$.

即曲率圆方程为: $(x-3)^2+(y+2)^2=8$.



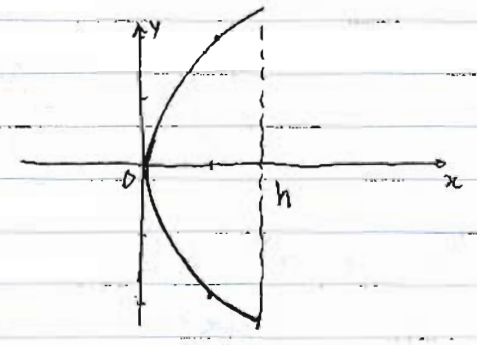
18. 求下列旋转曲面的面积.

(1) $y^2=4x$ $(0\le x\le h)$ 绕 x 旋转一周生成的曲面.

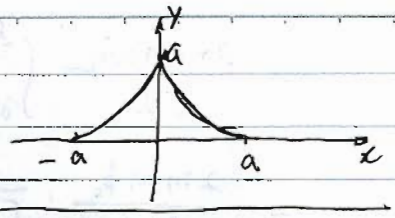
解: $y'=\frac{1}{\sqrt{x}}$.

$$F=2\pi\int_0^h\sqrt{4x}\cdot\sqrt{1+\frac{1}{x}}\,dx=4\pi\int_0^h\sqrt{x+1}\,dx.$$

$$=4\pi\cdot\frac{2}{3}(x+1)^{\frac{3}{2}}\Big|_0^h=\frac{8\pi}{3}\left[(h+1)^{3/2}-1\right]$$



(2) 星形线 $x=a\cos^3 t$, $y=a\sin^3 t$ 绕 x 轴一周生成的曲面.



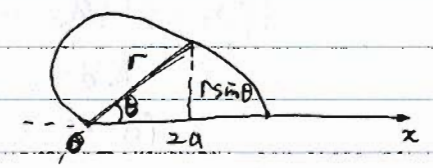
解: $\frac{dy}{dx}=\frac{3a\sin^2 t\cos t}{-3a\cos^2 t\sin t}=-\tan t$.

$$F=2\pi\int_0^{\pi} y(t)\cdot\sqrt{x'(t)^2+y'(t)^2}\,dt=2\pi\int_0^{\pi} a\sin^3 t\cdot\sqrt{9a^2(\cos^4 t\sin^2 t+\sin^4 t\cos^2 t)}\,dt$$

$$=6\pi a^2\int_0^{\pi}\sin^3 t\cdot|\sin t\cos t|\,dt=12\pi a^2\int_0^{\frac{\pi}{2}}\sin^4 t\cos t\,dt$$

$$=12\pi a^2\int_0^{\frac{\pi}{2}}\sin^4 t\,d\sin t=\frac{12}{5}\pi a^2\sin^5 t\Big|_0^{\frac{\pi}{2}}=\frac{12}{5}\pi a^2.$$

(3) 心脏线 $r=a(1+\cos\theta)$ $(a>0)$ 绕极轴旋转生成的曲面.



解: $r'=-a\sin\theta$. $r'^2=a^2\sin^2\theta$.

$$F=2\pi\int_0^{\pi} r\sin\theta\cdot\sqrt{r^2+r'^2}\,d\theta$$

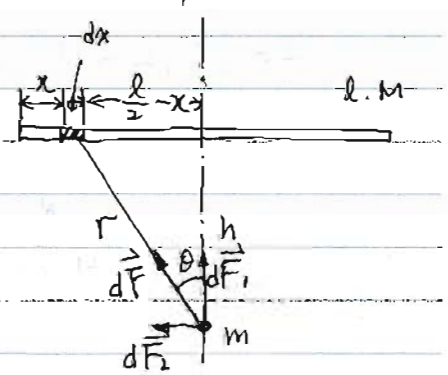
$$=2\pi\int_0^{\pi} a(1+\cos\theta)\cdot\sin\theta\cdot\sqrt{a^2(1+\cos\theta)^2+a^2\sin^2\theta}\,d\theta$$

$$=2\pi a^2\int_0^{\pi}(1+\cos\theta)\cdot\sin\theta\cdot\sqrt{2}\cdot\sqrt{1+\cos\theta}\,d\theta=-2\sqrt{2}\pi a^2\int_0^{\pi}(1+\cos\theta)^{\frac{3}{2}}d(1+\cos\theta)$$

$$=-2\sqrt{2}\pi a^2\cdot\frac{2}{5}(1+\cos\theta)^{\frac{5}{2}}\Big|_0^{\pi}=\frac{32}{5}\pi a^2.$$

19. 在牛顿引力场中，有一长为 l，质量为 M 的均匀细棒，在棒的垂直平分线上距棒 h 处有一质量为 m 的质点，求棒对此质点的引力.

解: 取棒上离端部 x 处的微元 dx. 微元质量 $dm=\frac{dx}{l}\cdot M$



$$|\vec{F}|=dF=k\cdot\frac{m\cdot dm}{r^2}\qquad dF_1=dF\cos\theta$$

水平方向的分量由对称性可知其互相抵消，合力为 0.

竖直方向合力:

$$F_\perp=\int_0^{l}d\vec{F}_1=\int_0^{l}dF\cos\theta$$

$$=\int_0^{l}k\cdot\frac{m\cdot\frac{dx}{l}\cdot M}{r^2}\cdot\frac{h}{r}=\frac{mMkh}{l}\int_0^{l}\frac{dx}{r^3}\qquad r=\sqrt{h^2+(\tfrac{l}{2}-x)^2}$$

$$=\frac{mMkh}{l}\int_0^{l}\left[h^2+(\tfrac{l}{2}-x)^2\right]^{-\frac{3}{2}}dx=\frac{2mMkh}{l}\int_0^{\frac{l}{2}}\frac{dx}{\left[h^2+(\frac{l}{2}-x)^2\right]^{\frac{3}{2}}}$$

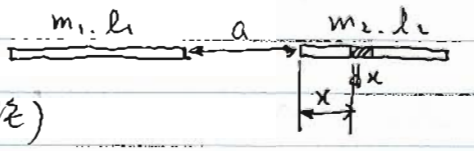
$$\left(\text{令 } t=x-\tfrac{l}{2}\right)=\frac{2mMkh}{l}\int_{-\frac{l}{2}}^{0}\frac{dt}{[h^2+t^2]^{\frac{3}{2}}}=\frac{2mMkh}{l}\int_0^{\frac{l}{2}}\frac{1}{t^3\left[1+\frac{h^2}{t^2}\right]^{\frac{3}{2}}}dt$$

$$= \frac{2mMkh}{h^2 \cdot l} \int_0^{\frac{l}{2}} \frac{-1}{\left[1+\frac{h^2}{t^2}\right]^{\frac{3}{2}}} d\left(\frac{h^2}{t^2}+1\right) = \frac{2mMk}{h \cdot l} \int_0^{\frac{l}{2}} \frac{-d\left(1+\frac{h^2}{t^2}\right)}{\left[1+\frac{h^2}{t^2}\right]^{\frac{3}{2}}}$$

$$= \frac{2mMk}{h \cdot l} \cdot \frac{1}{\sqrt{1+\frac{h^2}{t^2}}} \Bigg|_0^{\frac{l}{2}} = \frac{2mMk}{h\sqrt{4h^2+l^2}}.$$

20. 设有两均匀细杆长度分别为 l_1, l_2，质量分别为 m_1, m_2，它们位于同一条直线上，相邻两端距离为 a，试计算两杆间的引力。

解：l_2 上微元 dx，质量 $\frac{dx}{l_2} \cdot m_2 = \frac{m_2}{l_2} dx$.



l_1 杆对上述微元的引力为：（P.163. 例3.4.13结论）

$$dF = \frac{k\,dm_2 \cdot m_1}{(a+x)(l_1+a+x)} = \frac{km_1m_2}{l_2} \cdot \frac{dx}{(a+x)(l_1+a+x)}$$

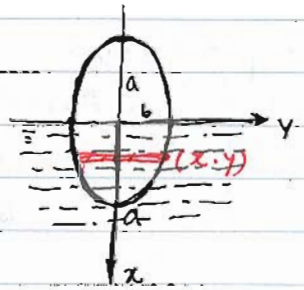
两杆间的引力：

$$F = \int_0^{l_2} dF = \int_0^{l_2} \frac{km_1m_2}{l_2} \frac{dx}{(a+x)(l_1+a+x)}$$

$$= \frac{km_1m_2}{l_1l_2} \int_0^{l_2} \left(\frac{1}{a+x} - \frac{1}{l_1+a+x}\right) dx = \frac{km_1m_2}{l_1l_2} \ln \frac{(l_1+a)(l_2+a)}{a(l_1+l_2+a)}$$

21. 有一椭圆形薄板，长半轴为 a，短半轴为 b，薄板垂直于水中，其短半轴与水平面相齐，求水对薄板的侧压力。

解：$dF = \rho g x \cdot 2y\,dx = 2\rho g x y\,dx$，$y = b\sqrt{1-\frac{x^2}{a^2}}$



$$F = \int_0^a dF = \int_0^a 2\rho g x \cdot b\sqrt{1-\frac{x^2}{a^2}}\,dx$$

$$= -\int_0^a a^2 \rho g b \sqrt{1-\frac{x^2}{a^2}}\, d\left(1-\frac{x^2}{a^2}\right)$$

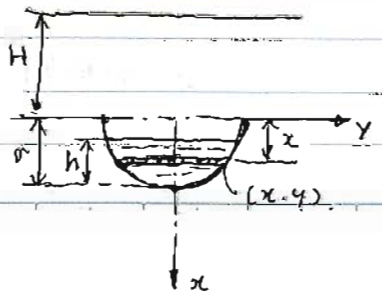
$$= -\frac{2}{3}\rho g a^2 b \left(1-\frac{x^2}{a^2}\right)^{\frac{3}{2}} \Bigg|_0^a = \frac{2}{3}\rho g a^2 b.$$

22. 数得水池的水抽到比水池高 H 的塔顶，若水池是半径为 r 的半球形，池中水深为 $h\,(h<r)$，求抽空这些水需作多少功？

解：微元 dx，其质量 $dm = \rho dV = \rho \pi y^2 dx$，而 $x^2+y^2=r^2$.

即：$dm = \pi\rho(r^2-x^2)dx$，$x: r-h \mapsto r$.

$$dW = dm \cdot g \cdot (x+H) = \pi\rho g(x+H)(r^2-x^2)dx.$$

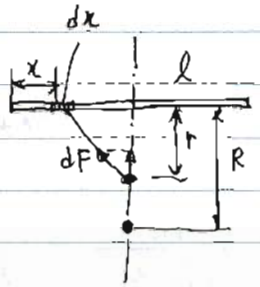


$$\therefore W=\int_{r-h}^{r} dW=\int_{r-h}^{r}\pi\rho g\,(x+H)(r^2-x^2)\,dx=\pi\rho g\int_{r-h}^{r}(Hr^2+r^2x-Hx^2-x^3)\,dx$$

$$=\pi\rho g\left(Hr^2x+\frac{1}{2}r^2x^2-\frac{1}{3}Hx^3-\frac{1}{4}x^4\right)\Big|_{r-h}^{r}$$

$$=\pi\rho g\left[Hr^2h+\frac{1}{2}r^2h(2r-h)-\frac{1}{3}Hh(3r^2-3rh+h^2)-\frac{1}{4}h(2r-h)(2r^2-2rh+h^2)\right].$$

23. 设有一均匀带正电细长为l的细杆，电荷线密度ρ为常数，在杆的垂直平分线上距杆为r处有一单位正电荷，(1)求带电细杆对该点电荷的斥力。(2)如该点电荷在杆的垂直平分线上从距杆r处移至距杆R处($R>r$)，求电场力所做的功。



解：(1) 微元dx，带电荷ρdx，到力微元dF为：

$$dF=\frac{k\cdot 1\cdot\rho\,dx}{r^2+(\frac{l}{2}-x)^2}$$ 由对称性知水平方向斥力为0.

$$dF_{\perp}=dF\cdot\frac{r}{\sqrt{r^2+(\frac{l}{2}-x)^2}}$$

$$\therefore F_{\perp}=\int_0^l dF_{\perp}=\int_0^l\frac{r}{\sqrt{r^2+(\frac{l}{2}-x)^2}}\cdot\frac{k\rho\,dx}{r^2+(\frac{l}{2}-x)^2}=rk\rho\int_0^l\frac{dx}{\left[r^2+(\frac{l}{2}-x)^2\right]^{3/2}}$$

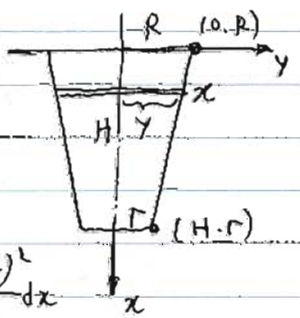
$$=2rk\rho\int_0^{\frac{l}{2}}\frac{dx}{\left[r^2+(\frac{l}{2}-x)^2\right]^{3/2}}=\frac{2k\rho l}{r\sqrt{4r^2+l^2}}$$ (积分过程同19题)

(2) $F_{\perp}(y)=\dfrac{2k\rho l}{y\sqrt{4y^2+l^2}}$ 变力作功：$W=\int\vec{F}\cdot d\vec{r}$

$$W=\int_r^R F_{\perp}(y)\,dy=\int_r^R\frac{2k\rho l}{y\sqrt{4y^2+l^2}}dy=-2k\rho\int_r^R\frac{d(\frac{l}{2y})}{\sqrt{1+(\frac{l}{2y})^2}}$$

$$=-2k\rho\ln\left|\frac{l}{2y}+\sqrt{1+\left(\frac{l}{2y}\right)^2}\right|\Big|_r^R=2k\rho\ln\left(\frac{R}{r}\cdot\frac{l+\sqrt{4r^2+l^2}}{l+\sqrt{4R^2+l^2}}\right).$$

24. 一盛满的半径为H米，顶口半径为R米，底口半径为r米，其中装满液体，试按流速公式$v=\mu\sqrt{2gh}$(μ为常数)，计算液体需多少时间流完。



解：设x处截面积$A(x)$，底口面积$\pi r^2=A_0$，则有：

$$A_0\cdot v\,dt=A(x)\cdot dx$$ (微段流出底口的通量)

$$dt=\frac{A(x)}{A_0\cdot v}dx,\quad A(x)=\pi y^2=\pi\left(R-\frac{R-r}{H}x\right)^2.$$

$$\therefore T=\int_0^H dt=\int_0^H\frac{\pi(R-\frac{R-r}{H}x)^2}{\pi r^2\cdot\mu\sqrt{2g(H-x)}}=\frac{1}{r^2\mu\sqrt{2g}}\int_0^H\frac{(R-\frac{R-r}{H}x)^2}{\sqrt{H-x}}dx$$

$$=\frac{\sqrt{2}}{r^2\mu\sqrt{g}}\int_0^H\left[r+\frac{R-r}{H}(H-x)\right]^2 d\sqrt{H-x}=\frac{\sqrt{2H}}{15\mu\sqrt{g}}\left(\frac{3R^2+4Rr+8r^2}{r^2}\right).$$

$y=R-\dfrac{R-r}{H}x.$

No. 110

Date

§5 广义积分

P.180. 习题

1. 计算下列收敛的广义积分：

(1) $\int_1^{+\infty} \frac{1}{(x+1)(x+2)}dx = \int_1^{+\infty}\left(\frac{1}{x+1}-\frac{1}{x+2}\right)dx = \ln\frac{x+1}{x+2}\Big|_1^{+\infty} = \ln\frac{1+\frac{1}{x}}{1+\frac{2}{x}}\Big|_1^{+\infty}$

$= -(\ln 2 - \ln 3) = \ln 3 - \ln 2.$

(2) $\int_0^{+\infty} e^{-x}\sin x\,dx = -\int_0^{+\infty}\sin x\,de^{-x} = -e^{-x}\sin x\Big|_0^{+\infty} + \int_0^{+\infty} e^{-x}\cos x\,dx$

$= -\int_0^{+\infty}\cos x\,de^{-x} = -e^{-x}\cos x\Big|_0^{+\infty} - \int_0^{+\infty} e^{-x}\sin x\,dx$

$= 1 - \int_0^{+\infty} e^{-x}\sin x\,dx.$

$\therefore \int_0^{+\infty} e^{-x}\sin x\,dx = \frac{1}{2}.$

(3) $\int_2^{+\infty}\frac{dx}{x(\ln x)^2} = \int_2^{+\infty}\frac{d(\ln x)}{(\ln x)^2} = -\frac{1}{\ln x}\Big|_2^{+\infty} = \frac{1}{\ln 2}.$

(4) $\int_0^{+\infty} e^{-\sqrt{x}}dx = 2\int_0^{+\infty}\sqrt{x}e^{-\sqrt{x}}d\sqrt{x} = 2\int_0^{+\infty} te^{-t}dt = -2\int_0^{+\infty} t\,de^{-t}$

$= -2te^{-t}\Big|_0^{+\infty} + 2\int_0^{+\infty} e^{-t}dt = \left(-2te^{-t}-2e^{-t}\right)\Big|_0^{+\infty} = -2.$

(5) $\int_1^{+\infty}\frac{\arctan x}{x^2}dx = -\int_1^{+\infty}\arctan x\,d\left(\frac{1}{x}\right) = -\frac{\arctan x}{x}\Big|_1^{+\infty} + \int_1^{+\infty}\frac{1}{x}\cdot\frac{1}{1+x^2}dx$

$= \frac{\pi}{4} + \int_1^{+\infty}\left(\frac{1}{x}-\frac{x}{1+x^2}\right)dx = \frac{\pi}{4} + \int_1^{+\infty}\frac{dx}{x} - \frac{1}{2}\int_1^{+\infty}\frac{d(1+x^2)}{1+x^2}$

$= \frac{\pi}{4} + \ln\frac{x}{\sqrt{1+x^2}}\Big|_1^{+\infty} = \frac{\pi}{4} + \ln\frac{1}{\sqrt{\frac{1}{x^2}+1}}\Big|_1^{+\infty} = \frac{\pi}{4} - \ln\frac{1}{\sqrt{2}} = \frac{\pi}{4} + \frac{1}{2}\ln 2.$

(6) $\int_0^{+\infty}\frac{dx}{1+x^3} = \int_0^{+\infty}\frac{dx}{(1+x)(1-x+x^2)} = \frac{1}{3}\int_0^{+\infty}\left(\frac{1}{1+x}-\frac{x-2}{1-x+x^2}\right)dx$

$= \frac{1}{3}\int_0^{+\infty}\left(\frac{dx}{1+x}\right) - \frac{1}{6}\int_0^{+\infty}\frac{2x-1}{x^2-x+1}dx + \frac{1}{2}\int_0^{+\infty}\frac{1}{x^2-x+1}dx$

$= \left[\frac{1}{3}\ln(1+x) - \frac{1}{6}\ln(x^2-x+1)\right]\Big|_0^{+\infty} + \frac{1}{2}\int_0^{+\infty}\frac{1}{(x-\frac{1}{2})^2+\frac{3}{4}}dx$

$= \frac{1}{6}\ln\frac{x^2+2x+1}{x^2-x+1}\Big|_0^{+\infty} + \frac{\sqrt{3}}{3}\arctan\frac{2x-1}{\sqrt{3}}\Big|_0^{+\infty}$

$= \frac{\sqrt{3}}{3}\left(\frac{\pi}{2}+\frac{\pi}{6}\right) = \frac{2\sqrt{3}}{9}\pi.$

(7) $\int_1^{+\infty} \frac{dx}{x\sqrt{1+x^5+x^{10}}}$ $\xrightarrow{\frac{1}{2}}$ $x^5=t$. $x: 1\to+\infty$, $t: 1\to+\infty$, $5x^4dx=dt$.

$$= \int_1^{+\infty} \frac{5x^4dx}{5x^5\sqrt{1+x^5+x^{10}}} = \int_1^{+\infty} \frac{dt}{5t\sqrt{1+t+t^2}} = \frac{1}{5}\int_1^{+\infty} \frac{dt}{t^2\sqrt{1+\frac{1}{t}+\frac{1}{t^2}}}$$

$$= -\frac{1}{5}\int_1^{+\infty} \frac{1}{\sqrt{(\frac{1}{t}+\frac{1}{2})^2+\frac{3}{4}}}\, d(\frac{1}{t}+\frac{1}{2}) = -\frac{1}{5}\ln\left|(\frac{1}{t}+\frac{1}{2})+\sqrt{1+\frac{1}{t}+\frac{1}{t^2}}\right|\Big|_1^{+\infty}$$

$$= -\frac{1}{5}\left[\ln\frac{3}{2} - \ln(\frac{3}{2}+\sqrt{3})\right] = \frac{1}{5}\ln\frac{2+\sqrt{3}}{\sqrt{3}} = \frac{1}{5}\ln(2+\sqrt{3}) - \frac{1}{10}\ln 3.$$

(8) $\int_0^{+\infty} \frac{x\ln x}{(1+x^2)^2}dx = \frac{1}{2}\int_0^{+\infty} \frac{\ln x}{(1+x^2)^2}d(1+x^2) = -\frac{1}{2}\int_0^{+\infty} \ln x\, d(\frac{1}{1+x^2})$

$$= -\frac{1}{2}\cdot\frac{\ln x}{1+x^2}\Big|_0^{+\infty} + \frac{1}{2}\int_0^{+\infty}\frac{1}{1+x^2}\cdot\frac{1}{x}dx$$

$$= -\frac{1}{2}\cdot\frac{\ln x}{1+x^2}\Big|_0^{+\infty} + \frac{1}{2}\int_0^{+\infty}\left(\frac{1}{x}-\frac{x}{1+x^2}\right)dx$$

$$= \left(-\frac{1}{2}\frac{\ln x}{1+x^2} + \frac{1}{2}\ln x\right)\Big|_0^{+\infty} - \frac{1}{4}\int_0^{+\infty}\frac{d(1+x^2)}{1+x^2}$$

$$= \left(-\frac{1}{2}\frac{\ln x}{1+x^2} + \frac{1}{2}\ln x - \frac{1}{4}\ln(1+x^2)\right)\Big|_0^{+\infty}$$

$$= -\frac{1}{2}\frac{\ln x}{1+x^2}\Big|_0^{+\infty} + \frac{1}{4}\ln\frac{x^2}{1+x^2}\Big|_0^{+\infty} = 0.$$

2. 判别下列无穷限广义积分的敛散性:

(1) $\int_1^{+\infty}\frac{\arctan x}{x^2}dx$.

解: $\lim_{x\to+\infty} x^2\cdot\left|\frac{\arctan x}{x^2}\right| = \lim_{x\to+\infty}|\arctan x| = \frac{\pi}{2} > 0$. 而 $p=2>1$.

∴ 原积分收敛 (柯西判别法)

(2) $\int_1^{+\infty}\frac{\cos^2 x}{x\sqrt{1+x}}dx$.

解: $\lim_{x\to+\infty} x^{\frac{3}{2}}\cdot\left|\frac{1}{x\sqrt{1+x}}\right| = \lim_{x\to+\infty}\sqrt{\frac{x}{1+x}} = 1 > 0$. 而 $p=\frac{3}{2}>0$.

∴ $\int_1^{+\infty}\frac{1}{x\sqrt{1+x}}dx$ 收敛. 而 $\frac{\cos^2 x}{x\sqrt{1+x}} \le \frac{1}{x\sqrt{1+x}}$, $x\in(1,+\infty)$.

∴ $\int_1^{+\infty}\frac{\cos^2 x}{x\sqrt{1+x}}dx$ 收敛. (比较判别法)

(3) $\int_{-\infty}^{-2} \frac{dx}{x+\sin x} = \int_{2}^{+\infty} \frac{dx}{x+\sin x}$.

解: $\lim_{x\to+\infty} x \cdot \frac{1}{x+\sin x} = \lim_{x\to+\infty} \frac{1}{1+\frac{\sin x}{x}} = 1 > 0$. 而 $p=1$.

$\therefore \int_{2}^{+\infty} \frac{dx}{x+\sin x}$ 发散. $\therefore$ 原积分发散.

(4) $\int_{1}^{+\infty} \frac{\ln^2 x}{(x+1)^2} dx$

解: $\lim_{x\to+\infty} x^{\frac{3}{2}} \cdot \frac{\ln^2 x}{(x+1)^2} = \lim_{x\to+\infty} \frac{\ln^2 x}{\sqrt{x}} \cdot \left(\frac{x}{x+1}\right)^2 = \lim_{x\to+\infty} \left(\frac{\ln x}{x^{\frac{1}{4}}}\right)^2 \cdot \left(\frac{x}{x+1}\right)^2$

$= 0 \cdot 1 = 0$. 而 $p = \frac{3}{2} > 1$.

$\therefore$ 原积分收敛.

(5) $\int_{1}^{+\infty} \arcsin\frac{1}{x^2} dx$.

解: $\lim_{x\to+\infty} x^2 \arcsin\frac{1}{x^2} = \lim_{x\to+\infty} \frac{\arcsin\frac{1}{x^2}}{\frac{1}{x^2}} \overset{令 t=\frac{1}{x^2}}{=\!=\!=} \lim_{t\to0^+} \frac{\arcsin t}{t}$

$= \lim_{t\to0^+} \frac{1}{\sqrt{1-t^2}} = 1 > 0$. 而 $p = 2 > 1$.

$\therefore$ 原积分收敛.

(6) $\int_{1}^{+\infty} \frac{\sqrt{1+x^{-1}} - 1}{x^3 \ln(1+x^2)} dx$.

解: $f(x) = \frac{\sqrt{1+x^{-1}} - 1}{x^3 \ln(1+x^2)} = \frac{\sqrt{1+x} - \sqrt{x}}{x^3\sqrt{x} \cdot \ln(1+x^2)} = \frac{1}{x^3\sqrt{x} \cdot (\sqrt{1+x} + \sqrt{x}) \cdot \ln(1+x^2)}$

$\lim_{x\to+\infty} x^3 \cdot f(x) = \lim_{x\to+\infty} \frac{1}{\sqrt{x} \cdot (\sqrt{1+x} + \sqrt{x}) \cdot \ln(1+x^2)} = 0$. 而 $p = 3 > 1$.

$\therefore$ 原积分收敛.

3. 计算下列广义积分:

(1) $\int_0^1 \ln x dx = x\ln x\Big|_0^1 - \int_0^1 x \cdot \frac{1}{x} dx = (x\ln x - x)\Big|_0^1 = -1 - \lim_{x\to0^+} x\ln x$

$= -1 - \lim_{x\to0^+} \frac{\ln x}{\frac{1}{x}} = -1 = \lim_{x\to0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = -1 + \lim_{x\to0^+} x = -1$.

(2) $\int_0^1 \frac{\arcsin x}{\sqrt{1-x^2}} dx = \int_0^1 \arcsin x \, d\arcsin x = \frac{1}{2}(\arcsin x)^2\Big|_0^1 = \frac{\pi^2}{8}$.

(3) $\int_0^1 \sqrt{x}\ln x\,dx = \frac{2}{3}\int_0^1 \ln x\,dx^{\frac{3}{2}} = \frac{2}{3}x^{\frac{3}{2}}\ln x\Big|_0^1 - \frac{2}{3}\int_0^1 x^{\frac{3}{2}}\cdot\frac{1}{x}dx$

$= \frac{2}{3}x^{\frac{3}{2}}\ln x\Big|_0^1 - \frac{2}{3}\cdot\frac{2}{3}x^{\frac{3}{2}}\Big|_0^1 = -\frac{4}{9} - \frac{2}{3}\lim_{x\to 0^+} x^{\frac{3}{2}}\ln x$

$= -\frac{4}{9} - \frac{2}{3}\lim_{x\to 0^+}\frac{\ln x}{x^{-\frac{3}{2}}} = -\frac{4}{9} - \frac{2}{3}\lim_{x\to 0^+}\frac{x^{-1}}{x^{-\frac{3}{2}}} = -\frac{4}{9}$.

(4) $\int_{-1}^1 \sqrt{\frac{1-x}{1+x}}\,dx$, 令 $\sqrt{\frac{1-x}{1+x}} = t$. 则 $x = \frac{1-t^2}{1+t^2}$, $dx = -\frac{4tdt}{(1+t^2)^2}$, $x: -1\to 1$, $t: +\infty\to 0$

$= \int_{+\infty}^0 t\cdot\frac{-4tdt}{(1+t^2)^2} = \int_0^{+\infty}\frac{4t^2dt}{(1+t^2)^2} = 4\int_0^{+\infty}\frac{dt}{1+t^2} - 4\int_0^{+\infty}\frac{dt}{(1+t^2)^2}$.

$= 4\arctan t\Big|_0^{+\infty} - 4\int_0^{\frac{\pi}{2}}\frac{\sec^2\theta\,d\theta}{\sec^4\theta}$ ($t=\tan\theta$, $t: 0\to+\infty$, $\theta: 0\to\frac{\pi}{2}$).

$= 4\cdot\frac{\pi}{2} - 4\int_0^{\frac{\pi}{2}}\cos^2\theta\,d\theta = 2\pi - 4\int_0^{\frac{\pi}{2}}\frac{1}{2}(1+\cos 2\theta)\,d\theta$

$= 2\pi - 2\left(\theta + \frac{1}{2}\sin 2\theta\right)\Big|_0^{\frac{\pi}{2}} = 2\pi - \pi = \pi$.

(5) $\int_a^b \frac{dx}{\sqrt{(x-a)(b-x)}}$ $(a<b)$.

解: 令 $x = a\cos^2 t + b\sin^2 t$. $x: a\to b$, $t: 0\to\frac{\pi}{2}$. $dx = (-2a\cos t\sin t + 2b\sin t\cos t)dt$

$= 2(b-a)\sin t\cdot\cos t\,dt$

$\int_a^b \frac{dx}{\sqrt{(x-a)(b-x)}} = \int_0^{\frac{\pi}{2}} \frac{2(b-a)\sin t\cdot\cos t\,dt}{\sqrt{(b\sin^2 t - a\sin^2 t)(b\cos^2 t - a\cos^2 t)}}$

$= \int_0^{\frac{\pi}{2}} \frac{2(b-a)\sin t\cdot\cos t\,dt}{(b-a)\sin t\cdot\cos t} = \int_0^{\frac{\pi}{2}} 2\,dt = 2t\Big|_0^{\frac{\pi}{2}} = \pi$.

4. 判别下列广义积分的收敛性:

(1) $\int_0^1 \frac{dx}{\sqrt{1-x^3}}$.

解: $\int_0^1 \frac{dx}{\sqrt{1-x^2}} = \arcsin x\Big|_0^1 = \frac{\pi}{2}$. 而 $\frac{1}{\sqrt{1-x^3}} < \frac{1}{\sqrt{1-x^2}}$, $x\in(0,1)$.

∴ 原积分收敛.

(2) $\int_1^2 \frac{2}{\ln^2 x}dx$

解: $\lim_{x\to 1+0}(x-1)^2\frac{2}{\ln^2 x} = 2\lim_{x\to 1+0}\frac{(x-1)^2}{\ln^2 x} = 2\lim_{x\to 1+0}\frac{2(x-1)}{2\ln x\cdot\frac{1}{x}} = 2\lim_{x\to 1+0}\frac{x^2-x}{\ln x}$

$= 2\lim_{x\to 1+0}\frac{2x}{\frac{1}{x}} = 4\lim_{x\to 1+0}x^2 = 4 > 0$, 而 $p = 2 > 1$.

∴ 原积分发散.

(3) $\int_0^{+\infty} \frac{dx}{\sqrt{x}(1+x)}$.

解: $\lim_{x\to+\infty} x^{\frac{3}{2}} \cdot f(x) = \lim_{x\to+\infty} \frac{x}{1+x} = 1$. 而 $p=\frac{3}{2}>1$. $\therefore \int_a^{+\infty} \frac{dx}{\sqrt{x}(1+x)}$ 收敛.

$\lim_{x\to 0+0} x^{\frac{1}{2}} \cdot f(x) = \lim_{x\to 0+0} \frac{1}{1+x} = 1$. 而 $p=\frac{1}{2}<1$. $\therefore \int_0^a \frac{dx}{\sqrt{x}(1+x)}$ 收敛.

$\therefore$ 原积分收敛

(4) $\int_0^{+\infty} \frac{\ln x}{x^2} dx$.

解: $\lim_{x\to+\infty} x^{\frac{3}{2}} \cdot f(x) = \lim_{x\to+\infty} \frac{\ln x}{\sqrt{x}} = \lim_{x\to+\infty} \frac{\frac{1}{x}}{\frac{1}{2}x^{-\frac{1}{2}}} = \lim_{x\to+\infty} \frac{2}{\sqrt{x}} = 0$. 而 $p=\frac{3}{2}>1$.

$\therefore \int_a^{+\infty} \frac{\ln x}{x^2} dx$ 收敛.

$\lim_{x\to 0+0} x^2 \cdot |f(x)| = \lim_{x\to 0+0} x^2 \cdot \frac{|\ln x|}{x^2} = \lim_{x\to 0+0} |\ln x| = +\infty$. 而 $p=2>1$.

$\therefore \int_0^a \frac{\ln x}{x^2} dx$ 发散.

$\therefore \int_0^{+\infty} \frac{\ln x}{x^2} dx = \int_0^a \frac{\ln x}{x^2} dx + \int_a^{+\infty} \frac{\ln x}{x^2} dx$. 发散.

(5) $\int_0^{+\infty} \frac{\sin x}{\sqrt{x^3}} dx$.

解: $\lim_{x\to+\infty} x^{\frac{5}{4}} \cdot |f(x)| = \lim_{x\to+\infty} x^{\frac{5}{4}} \cdot \frac{|\sin x|}{x^{\frac{3}{2}}} = \lim_{x\to+\infty} \left|\frac{\sin x}{x^{\frac{1}{4}}}\right| = 0$. 而 $p=\frac{5}{4}>1$.

$\therefore \int_a^{+\infty} \frac{\sin x}{\sqrt{x^3}} dx$ 收敛.

$\lim_{x\to 0+0} x^{\frac{1}{2}} \cdot |f(x)| = \lim_{x\to 0+0} x^{\frac{1}{2}} \cdot \frac{|\sin x|}{x^{\frac{1}{2}}} = \lim_{x\to 0+0} \left|\frac{\sin x}{x}\right| = 1 > 0$. 而 $p=\frac{1}{2}<1$.

$\therefore \int_0^a \frac{\sin x}{\sqrt{x^3}} dx$ 收敛.

$\therefore \int_0^{+\infty} \frac{\sin x}{\sqrt{x^3}} dx = \int_0^a \frac{\sin x}{\sqrt{x^3}} dx + \int_a^{+\infty} \frac{\sin x}{\sqrt{x^3}} dx$ 收敛.

(6) $\int_0^{+\infty} \frac{\ln(1+x)}{x^p} dx$

解: $\int_0^{+\infty} \frac{\ln(1+x)}{x^p} dx = \int_0^1 \frac{\ln(1+x)}{x^p} dx + \int_1^{+\infty} \frac{\ln(1+x)}{x^p} dx$.

$\frac{\ln(1+x)}{x^p} \sim \frac{x}{x^p} = \frac{1}{x^{p-1}}$ $(x \to 0+0)$.

∴ 当 $p<2$ 时. $\int_0^1 \frac{\ln(1+x)}{x^p}dx$ 收敛.

当 $p \geq 2$ 时. $\int_0^1 \frac{\ln(1+x)}{x^p}dx$ 发散.

当 $p>1$ 时. $\lim_{x\to+\infty} x^{\frac{p+1}{2}} \cdot \frac{\ln(1+x)}{x^p} = \lim_{x\to+\infty} \frac{\ln(1+x)}{x^{\frac{p-1}{2}}} = 0$. 而 $\frac{p+1}{2}>1$. $(p>1)$.

∴ 此时 $\int_1^{+\infty} \frac{\ln(1+x)}{x^p}dx$ 收敛.

当 $p \leq 1$ 时. $\lim_{x\to+\infty} x^{\frac{p+1}{2}} \cdot \frac{\ln(1+x)}{x^p} = \lim_{x\to+\infty} \frac{\ln(1+x)}{x^{\frac{p-1}{2}}} = +\infty$. 而 $\frac{p+1}{2} \leq 1$. $(p \leq 1)$.

∴ 此时 $\int_1^{+\infty} \frac{\ln(1+x)}{x^p}dx$ 发散.

综上所述. 当 $1<p<2$ 时. $\int_0^{+\infty} \frac{\ln(1+x)}{x^p}dx$ 收敛.

5. 计算下列 Cauchy 积分积分:

(1) $(cpV)\int_{\frac{1}{2}}^{2} \frac{dx}{x\ln x} = (cpV)\int_{\frac{1}{2}}^{2} \frac{1}{\ln x}d(\ln x) = (cpV)\ln|\ln x|\Big|_{\frac{1}{2}}^{2}$

$= \lim_{\varepsilon\to 0}\left(\ln|\ln x|\Big|_{\frac{1}{2}}^{1-\varepsilon} + \ln|\ln x|\Big|_{1+\varepsilon}^{2}\right) = \lim_{\varepsilon\to 0}\left(\ln|\ln(1-\varepsilon)| - \ln|\ln\frac{1}{2}| + \ln|\ln(1+\varepsilon)|\right)$

$+\ln\ln 2 - \ln|\ln\frac{1}{2}| = \lim_{\varepsilon\to 0}\ln\frac{\ln(1-\varepsilon)}{\ln(1+\varepsilon)} + \ln\ln 2 - \ln\ln 2$

$= 0$.

(2) $(cpV)\int_0^{+\infty} \frac{dx}{x^2-3x+2} = (cpV)\int_0^{+\infty}\left(\frac{1}{x-2} - \frac{1}{x-1}\right)dx = (cpV)\ln\left|\frac{x-2}{x-1}\right|\Big|_0^{+\infty}$

$(\varepsilon\to 0)$ $= \ln\left|\frac{1-\frac{2}{x}}{1-\frac{1}{x}}\right|\Big|_{3+\varepsilon}^{+\infty} + \ln\left|\frac{x-2}{x-1}\right|\Big|_{3+\varepsilon}^{2-\varepsilon} + \ln\left|\frac{x-2}{x-1}\right|\Big|_{0}^{1-\varepsilon}$

$(\varepsilon\to 0)$ $= -\ln\left|\frac{-2-3+2}{-2-3+1}\right| + \ln\left|\frac{1-3-2}{-2-3-2}\right| - \ln\left|\frac{1-3+1}{-2-3+1}\right| + \ln\left|\frac{1-3-1}{-2-3-1}\right| - \ln 2$

$(\varepsilon\to 0)$ $= -\ln|3+1| - \ln|3| + \ln|3-1| + \ln|3-2| + \ln|3+1| + \ln|3| - \ln|3| - \ln 2$

$= 2\lim_{\varepsilon\to 0}\ln\left|\frac{3+1}{3-1}\right| - \ln 2 = -\ln 2$.

6. 把下列积分表示为 Γ 函数.

(1) $\int_0^{+\infty} e^{-x^n}dx$ $(n>0)$.

解: 设 $x^n=t$, 则 $x=t^{\frac{1}{n}}$, $dx=\frac{1}{n}t^{\frac{1}{n}-1}dt$. $x:\ 0\mapsto+\infty$, $t:\ 0\mapsto+\infty$.

$$\int_0^{+\infty} e^{-x^n}dx=\int_0^{+\infty} e^{-t}\cdot\frac{1}{n}t^{\frac{1}{n}-1}dt=\frac{1}{n}\int_0^{+\infty} t^{\frac{1}{n}-1}e^{-t}dt=\frac{\Gamma(\frac{1}{n})}{n}.$$

(2) $\int_0^{+\infty} x^m e^{-x^n}dx$.

解: 设 $x^n=t$, 则 $x=t^{\frac{1}{n}}$, $dx=\frac{1}{n}t^{\frac{1}{n}-1}dt$. $x:\ 0\mapsto+\infty$, $t:\ 0\mapsto+\infty$.

$$\int_0^{+\infty} x^m e^{-x^n}dx=\int_0^{+\infty} t^{\frac{m}{n}}\cdot e^{-t}\cdot\frac{1}{n}t^{\frac{1}{n}-1}dt=\frac{1}{n}\int_0^{+\infty} t^{\frac{m+1}{n}-1}e^{-t}dt=\frac{\Gamma(\frac{m+1}{n})}{n}.$$

(3) $\int_0^1 (\ln\frac{1}{x})^p dx$.

解: 设 $\ln\frac{1}{x}=t$, 则 $x=e^{-t}$, $dx=-e^{-t}dt$. $x:\ 0\mapsto1$, $t:\ +\infty\mapsto0$.

$$\int_0^1 (\ln\frac{1}{x})^p dx=\int_{+\infty}^0 t^p\cdot(-e^{-t})dt=\int_0^{+\infty} t^{p+1-1}\cdot e^{-t}dt=\Gamma(p+1).$$

7. 求 $\Gamma(\frac{2n+1}{2})$.

解: $\Gamma(\frac{2n+1}{2})=\Gamma(n+\frac{1}{2})=(n-\frac{1}{2})\Gamma(n-\frac{1}{2})=(n-\frac{1}{2})(n-\frac{3}{2})\cdots\frac{3}{2}\cdot\frac{1}{2}\Gamma(\frac{1}{2})$

$$=\frac{(2n-1)(2n-3)\cdots3\cdot1}{2^n}\cdot\Gamma(\frac{1}{2})=\frac{(2n-1)!!}{2^n}\cdot\sqrt{\pi}.$$

8. 利用 Γ 函数计算积分:

(1) $\int_0^{+\infty} x^{\frac{3}{2}}e^{-4x}dx=4^{-\frac{5}{2}}\int_0^{+\infty}(4x)^{\frac{5}{2}-1}e^{-4x}d4x=\frac{1}{32}\int_0^{+\infty}t^{\frac{5}{2}-1}e^{-t}dt$

$$=\frac{1}{32}\Gamma(\frac{5}{2})=\frac{1}{32}\cdot\frac{3}{2}\cdot\frac{1}{2}\cdot\sqrt{\pi}=\frac{3}{128}\sqrt{\pi}.$$

(2) $\int_0^{+\infty} x^{2n}e^{-x^2}dx=\frac{1}{2}\int_0^{+\infty}(x^2)^{n-\frac{1}{2}}e^{-x^2}dx^2=\frac{1}{2}\int_0^{+\infty}t^{n-\frac{1}{2}}e^{-t}dt$

$$=\frac{1}{2}\int_0^{+\infty}t^{n+\frac{1}{2}-1}e^{-t}dt=\frac{1}{2}\Gamma(\frac{2n+1}{2})=\frac{(2n-1)!!}{2^{n+1}}\sqrt{\pi}.$$

9. 证明: $B(p,q)\,B(p+q,r) = B(q,r)\,B(q+r,p)$

证明:

$$B(p,q) = \int_0^1 x^{p-1}(1-x)^{q-1}\,dx = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}$$

$$B(p+q,r) = \int_0^1 x^{p+q-1}(1-x)^{r-1}\,dx = \frac{\Gamma(p+q)\cdot\Gamma(r)}{\Gamma(p+q+r)}$$

$$B(q,r) = \int_0^1 x^{q-1}(1-x)^{r}\,dx = \frac{\Gamma(q)\cdot\Gamma(r)}{\Gamma(q+r)}$$

$$B(q+r,p) = \int_0^1 x^{q+r-1}(1-x)^{p}\,dx = \frac{\Gamma(q+r)\Gamma(p)}{\Gamma(p+q+r)}$$

$$B(p,q)\,B(p+q,r) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}\cdot\frac{\Gamma(p+q)\cdot\Gamma(r)}{\Gamma(p+q+r)} = \frac{\Gamma(p)\cdot\Gamma(q)\cdot\Gamma(r)}{\Gamma(p+q+r)}$$

$$B(q,r)\,B(q+r,p) = \frac{\Gamma(q)\cdot\Gamma(r)}{\Gamma(q+r)}\cdot\frac{\Gamma(q+r)\cdot\Gamma(p)}{\Gamma(p+q+r)} = \frac{\Gamma(p)\cdot\Gamma(q)\cdot\Gamma(r)}{\Gamma(p+q+r)}$$

$\therefore B(p,q)\,B(p+q,r) = B(q,r)\cdot B(q+r,p)$.

10. 证明: 当 $a>0$ 时, 只要下式两边的反常积分有意义, 则有:

$$\int_0^{+\infty} f\left(\frac{x}{a}+\frac{a}{x}\right)\frac{\ln x}{x}\,dx = \ln a\int_0^{+\infty} f\left(\frac{x}{a}+\frac{a}{x}\right)\frac{1}{x}\,dx.$$

证明: 令 $x=\frac{a^2}{t}$, 则 $dx = -\frac{a^2}{t^2}dt$. $x: a\to+\infty$, $t: a\to 0$. 且 $\frac{x}{a}+\frac{a}{x} = \frac{t}{a}+\frac{a}{t}$.

$$\int_0^{+\infty} f\left(\frac{x}{a}+\frac{a}{x}\right)\frac{\ln x}{x}\,dx = \int_0^{a} f\left(\frac{x}{a}+\frac{a}{x}\right)\frac{\ln x}{x}dx + \int_a^{+\infty} f\left(\frac{x}{a}+\frac{a}{x}\right)\frac{\ln x}{x}\,dx.$$

(x 换成 t)

$$= \int_0^a f\left(\frac{x}{a}+\frac{a}{x}\right)\frac{\ln x}{x}\,dx + \int_a^0 f\left(\frac{t}{a}+\frac{a}{t}\right)\frac{2\ln a-\ln t}{a^2}\cdot t\cdot\left(-\frac{a^2}{t^2}\right)dt$$

$$= \int_0^a f\left(\frac{x}{a}+\frac{a}{x}\right)\frac{\ln x}{x}\,dx + \int_0^a f\left(\frac{t}{a}+\frac{a}{t}\right)\frac{2\ln a-\ln t}{t}\,dt$$

$$= \int_0^a f\left(\frac{x}{a}+\frac{a}{x}\right)\frac{\ln x}{x}\,dx + 2\ln a\int_0^a f\left(\frac{x}{a}+\frac{a}{x}\right)\cdot\frac{1}{x}\,dx - \int_0^a f\left(\frac{x}{a}+\frac{a}{x}\right)\frac{\ln x}{x}dx$$

$$= 2\ln a\int_0^a f\left(\frac{x}{a}+\frac{a}{x}\right)\cdot\frac{1}{x}\,dx = \ln a\int_0^a f\left(\frac{x}{a}+\frac{a}{x}\right)\cdot\frac{1}{x}\,dx + \ln a\int_0^a f\left(\frac{a}{t}+\frac{a}{t}\right)\cdot\frac{1}{t}\,dt$$

(t 换成 x)

$$= \ln a\int_0^a f\left(\frac{x}{a}+\frac{a}{x}\right)\cdot\frac{1}{x}\,dx + \ln a\int_{+\infty}^a f\left(\frac{x}{a}+\frac{a}{x}\right)\cdot\frac{x}{a^2}\cdot\left(-\frac{(a^2)^2}{a^2}\right)dx$$

$$= \ln a\int_0^a f\left(\frac{x}{a}+\frac{a}{x}\right)\cdot\frac{1}{x}\,dx + \ln a\int_a^{+\infty} f\left(\frac{x}{a}+\frac{a}{x}\right)\cdot\frac{1}{x}\,dx$$

$$= \ln a\int_0^{+\infty} f\left(\frac{x}{a}+\frac{a}{x}\right)\cdot\frac{1}{x}\,dx.$$